补充材料：NNS 计算图解释与 ViT 结构路径分析

返回正文。本文件包含复现参数、完整统计、历史图索引与后续实验范围。主文 RQ 编号按修订版使用。

S1. Model and registry details

Models: `vit_base_patch16_224.augreg2_in21k_ft_in1k` and `deit_small_patch16_224.fb_in1k`. Torch 2.2.2+cu121; timm 0.9.16; torchvision 0.17.2+cu121; NumPy 1.26.4; Pillow 10.2.0.

Table S1. Original scalar registry coverage.

Model	Object	Candidate definition	Candidates	Registered/image	Coverage	Strategy	Scalar neuron
vit_b16	post-block token-channel	12*197*D	1,815,552	192	0.01058%	12 blocks x 4 tokens x 4 channels	yes
vit_b16	post-block patch sum/energy	2*12*D	18,432	96	0.52083%	12 blocks x 4 channels x 2 aggregates	no
vit_b16	attention-head coordinate	12*H*197*64	1,815,552	144	0.00793%	all heads; CLS token; within-head channel 0	yes
vit_b16	module patch aggregate	3*(4D+24*D)	27,648	18	0.06510%	blocks 0/5/11; six modules; channel 0	no
vit_b16	patch embedding aggregate	D	768	4	0.52083%	four fixed channels; sum over patches	no
vit_b16	final CLS coordinate	D	768	1	0.13021%	channel 0	yes
vit_b16	ground-truth logit	1000	1,000	1	0.10000%	frozen image label	output
vit_b16	clean-top1 logit	1000	1,000	1	0.10000%	frozen clean prediction	output
deit_s16	post-block token-channel	12*197*D	907,776	192	0.02115%	12 blocks x 4 tokens x 4 channels	yes
deit_s16	post-block patch sum/energy	2*12*D	9,216	96	1.04167%	12 blocks x 4 channels x 2 aggregates	no
deit_s16	attention-head coordinate	12*H*197*64	907,776	72	0.00793%	all heads; CLS token; within-head channel 0	yes
deit_s16	module patch aggregate	3*(4D+24*D)	13,824	18	0.13021%	blocks 0/5/11; six modules; channel 0	no

Model	Object	Candidate definition	Candidates	Registered/image	Coverage	Strategy	Scalar neuron
deit_s16	patch embedding aggregate	D	384	4	1.04167%	four fixed channels; sum over patches	no
deit_s16	final CLS coordinate	D	384	1	0.26042%	channel 0	yes
deit_s16	ground-truth logit	1000	1,000	1	0.10000%	frozen image label	output
deit_s16	clean-top1 logit	1000	1,000	1	0.10000%	frozen clean prediction	output

Table S2. Resources and extended validation coverage.

Resource / validation	Value
Formal scan wall time	1,752.71 s
Four-card GPU-hours upper bound	1.9475
Throughput	192.16 targets/s
Retained data	9.290 GB
Peak reserved GPU memory	2.284 GB
Peak CPU RSS	2.018 GB
M1 random permutations	128,000,000
M4 attention / residual / JVP records	86,400 / 19,200 / 691,200

S1.1 实现与核心指标伪代码

模型均处于 evaluation mode。M3 使用 torch.autograd.grad 的 is_grads_batched 模式分块执行 VJP，冻结 gradient_chunk_size=16；具体随机种子、模型与图像哈希、worker 环境和每任务耗时记录在对应 evidence JSON 中。主扫描的时间、吞吐、显存与保留数据量见 Table S2；补充控制实验的 456 个任务合计 2.4502 GPU-hours。核心实现位于 src/vitnns/endpoint_m3_atlas.py、src/vitnns/endpoint_m4.py 与相应 scripts/run_endpoint_*.py。

```
Endpoint(s, x):
    g <- VJP(s, x) at the observed forward state
    return x * g

StructuredCut(F, edges, reconstruction_target):
    for each actual use-edge z_k in a valid cut:
        g_k <- downstream cotangent dF/dz_k
        E_k <- x * VJP(z_k, x; vector=g_k)
    assert sum_k E_k closes to reconstruction_target # branch contribution or Total Endpoint

SpatialMetrics(E, target_patch):
    M[p] <- sum_rgb abs(E[p, rgb])
    if sum_p M[p] == 0: return NA
    p_patch <- patch_sum(M) / sum_p M[p]
    entropy <- -sum_j p_patch[j] log(p_patch[j]) / log(196)
    locality <- neighborhood_mass(target_patch) - uniform_neighborhood_mass
    distance <- sum_j p_patch[j] normalized_distance(j, target_patch)
```

```
RandomRegionAUC(repeats):
    for each random repeat r:
        curve_r <- absolute finite effect at every area budget
        auc_r <- trapezoid(curve_r, area in [0, 0.25]) / 0.25
    return mean_r auc_r
```

CLS 没有自然的目标 patch，因此 locality 与 target-relative distance 仅对 patch-token 目标定义；零质量样本传播为 NA。图内 token、channel 与 head 先按预定规则汇总，再以图像为统计单位。

S2. Frozen decisions and numerical diagnostics

C4a: BROAD_BUT_NON_UNIVERSAL_FAMILY_LEVEL_SUPPORT; original all-family omnibus NOT_SUPPORTED. C4b: NOT_SUPPORTED. The numerical atlas status is PASS_FLOAT64_RECOVERY_WITH_RETAINED_FLOAT32_FAI 36/57,600 float32 failures are retained and 36/36 same-coordinate float64 checks pass. These fields are not interchangeable.

Table S2.1. 原 pooled C4a provenance；图像级中位数 [95% CI]。 Holm p 来自原成分和区域合计 28 项的冻结 family。

Model	Reference	Heads	A/V pooled	Residual pooled
vit_b16	normalized-zero	0.291 [0.281, 0.298]	0.0996 [0.075, 0.123]	0.177 [0.155, 0.195]
vit_b16	blur	0.306 [0.296, 0.318]	0.157 [0.131, 0.177]	0.209 [0.182, 0.237]
deit_s16	normalized-zero	0.191 [0.177, 0.205]	0.107 [0.078, 0.129]	0.135 [0.113, 0.164]
deit_s16	blur	0.182 [0.168, 0.195]	0.160 [0.128, 0.185]	0.229 [0.204, 0.251]

12 项原 Holm 调整后 p 均小于 10^{-6}。原 composite 中 11/12 family 达到操作门槛，但 all-family C4a 仍为 NOT_SUPPORTED。这张表保存历史判定身份；S18 的分层诊断不追溯改变它。

S3. Reference sensitivity

Change uses atol=1e-12 and rtol=0. Relative change is abs(q0-qblur)/max(abs(q0),abs(qblur)); both-zero denominators are NA. Component quantiles describe the pooled records, not independent inferential samples.

Table S3. Changes in Taylor magnitude and absolute finite effect.

Model	Quantity	Component pairs	Above tolerance	Abs change median [q10,q90]	Relative change median [q10,q90]
DeiT-S/16	finite_effect	67200	100.0%	0.1365 [0.0161, 1.6575]	0.4927 [0.0985, 0.8997]
DeiT-S/16	taylor	67200	100.0%	0.1215 [0.0150, 0.7867]	0.5186 [0.1172, 0.9033]
ViT-B/16	finite_effect	96000	100.0%	0.0713 [0.0076, 1.4414]	0.5300 [0.1117, 0.9137]
ViT-B/16	taylor	96000	100.0%	0.0688 [0.0078, 0.5471]	0.5225 [0.1133, 0.9052]

Head rankings use average ranks and Kendall tau-b. Top-K ties are broken by ascending component ID; Top-3 overlap is intersection size / 3. Rank displacement uses original rank positions, so model head counts affect its scale. Each image averages 12 blocks, then summaries use 400 images. Full-order equality counts are Taylor 0/4,800 and effect 0/4,800 for ViT-B, Taylor 46/4,800 and effect 64/4,800 for DeiT-S.

Table S4. Image-level rank concordance distributions.

Model	Quantity	Metric	Image median [q10,q90]	n / NA
DeiT-S/16	finite_effect	spearman	0.3071 [0.1238, 0.4524]	400 / 0
DeiT-S/16	finite_effect	kendall_tau_b	0.2444 [0.0989, 0.3678]	400 / 0
DeiT-S/16	finite_effect	top1_agreement	0.4167 [0.1667, 0.5833]	400 / 0
DeiT-S/16	finite_effect	top3_overlap	0.6111 [0.5278, 0.6944]	400 / 0
DeiT-S/16	finite_effect	mean_absolute_rank_displacement	1.5278 [1.2778, 1.7528]	400 / 0
DeiT-S/16	taylor	spearman	0.3000 [0.1000, 0.4476]	400 / 0
DeiT-S/16	taylor	kendall_tau_b	0.2389 [0.0778, 0.3667]	400 / 0
DeiT-S/16	taylor	top1_agreement	0.3333 [0.1667, 0.5833]	400 / 0
DeiT-S/16	taylor	top3_overlap	0.6111 [0.5278, 0.6944]	400 / 0
DeiT-S/16	taylor	mean_absolute_rank_displacement	1.5278 [1.3056, 1.8056]	400 / 0
ViT-B/16	finite_effect	spearman	0.3759 [0.2622, 0.4904]	400 / 0
ViT-B/16	finite_effect	kendall_tau_b	0.2803 [0.1967, 0.3740]	400 / 0
ViT-B/16	finite_effect	top1_agreement	0.3333 [0.2500, 0.5833]	400 / 0
ViT-B/16	finite_effect	top3_overlap	0.5000 [0.4167, 0.6111]	400 / 0
ViT-B/16	finite_effect	mean_absolute_rank_displacement	2.9444 [2.6111, 3.2917]	400 / 0
ViT-B/16	taylor	spearman	0.3782 [0.2539, 0.4914]	400 / 0
ViT-B/16	taylor	kendall_tau_b	0.2828 [0.1942, 0.3737]	400 / 0
ViT-B/16	taylor	top1_agreement	0.3333 [0.2500, 0.5833]	400 / 0
ViT-B/16	taylor	top3_overlap	0.5000 [0.4167, 0.6111]	400 / 0
ViT-B/16	taylor	mean_absolute_rank_displacement	2.9444 [2.5972, 3.2792]	400 / 0

S4. Method objects

Table S5. Functional comparison.

Method	Primary object	External reference	Propagation / contract
Endpoint	Normalized-input Total Endpoint	No; normalization origin matters	Global cuts close to total; branch partitions close to branch
Activation-space Taylor	Specified replacement effect	Yes	Clean cotangent × displacement
Integrated Gradients	Baseline-to-input attribution	Yes	Path-average gradient; baseline-difference completeness
Attention rollout	Token flow	No	Accumulated attention / residual flow
Grad-CAM	Class-conditioned coarse localization	No	Gradient-weighted feature maps
Chefer / AttnLRP	Transformer relevance	Implementation-specific	Method-specific relevance rules
Raw attention	Token allocation	No	Forward attention weights

Table S5a. 完整近邻差分矩阵。“未报告” 限于参考文献表所列核验版本；不同方法的 closure 对象不可互换。

Method family	Explained object	Consumer edge/use	Spatial component map	Closure object	Finite intervention
Gradient × Input	当前点输入加权梯度	未报告	单一输入图	一般不满足函数值完整性	否
Integrated Gradients	reference-to-input 路径归因	未报告	单一输入图	输出差	否
Conductance	经过隐藏单元的路径归因	隐藏单元集合	可选输入分辨率	层级输出差	否
GlobEnc / ALTI	跨层 token contribution/mixing	算子级	token 图	局部前向分解	否
DecompX	representation 与 logit 的 token components	组件级	token 分项	依 bias 分配规则重组	否
AttnLRP / HiLRP / GradSkip	Transformer relevance	算子连接	relevance 图	relevance conservation	否
EAP / AtP* / Vi-CD	reference-specific component/edge effect	是	通常不保留本文的输入坐标分项	patching effect 或 circuit fidelity	是

Method family	Explained object	Consumer edge/use	Spatial component map	Closure object	Finite intervention
Activation patching / CAAP	内部 activation 替换效应	node 或指定 activation	patch score	一般不加和	是
DAVE	ViT input-gradient 分解	未报告 consumer-use 索引	像素图	effective input-output map	否
Unpack	component interaction、命名路径与 token credit	组件/路径身份	token attribution	方法特定 credit decomposition	否
NNS–ViT	指定 use 与输入位置的同一当前点局部微分	consumer、slot、invocation	分项输入图	声明 branch 或 global cut 的同一 signed differential	分开评价

S4.1 中英术语合同

中文	English	定义边界
结构合法切面	structurally valid cut	每条输入到目标的结构 use-path 恰穿一次，与当前点数值无关
点态微分闭合	pointwise differential closure	固定输入处 use-specific Jacobian/cotangent 分组之和
分支微分分组	branch differential partition	仅重构 attention 或 MLP branch contribution
全局切面	global cut	包含 bypass 并重构同目标 Total Endpoint
路径图	path map	指定 tensor use 的局部微分分项，不是唯一因果路径
局部微分绝对质量比	local-differential absolute-mass ratio	routing/identity scale；不是因果贡献率
区域诱导的指定-head activation 替换	region-induced replacement of designated head activations	混合输入产生 head activation，再插入 clean forward 的两阶段任务
候选集合遗憾率	candidate-set regret	相对已评价候选集合包络，不是全局最优策略 regret

S5. Corrected absolute random AUC and oracle

当前随机基线对每次重复先取绝对值并积分，再平均 AUC。修正结果、重复分布和 MC 稳定性见supplementary experiments S9. 旧分析作为原估计器的历史复核保留。

S6. Detailed attention/residual evidence (historical stage identifiers retained)

历史 M4：Attention routing/value 与 residual 机制

M4 完成两模型共享 400 图、12 个 blocks 和全部 heads 的 ground-truth-logit 局部输入依赖分解，共 86,400 个 image-block-head attention records、19,200 个 image-block-residual records 和 691,200 个 direction-head JVP records。Q+K+V 逐 head 重构 native attention Endpoint；routing+value 重构同一对象；attention 和 MLP residual 的 identity/carry 加 branch 均重构完整 logit Endpoint。

独立 CPU 审计复算 230,400 个 E3 summary 字段。最大 Q/K/V closure 绝对误差为 1.574e-05，最大 residual closure 绝对误差为 2.480e-05，显式 attention differential 与 native JVP 的最大相对误差为 2.502e-05。全部 19,200 个省略 identity/carry 负控按预期闭合失败；冻结前 16 个 evaluation orders 的 384 个 attention-detach block checks 通过。

Attention routing/value

模型	Block	Routing share	Frozen-attention Endpoint error	Frozen-attention JVP error	Mean-query saturation	Routing-value path cancellation
vit_b16	0	0.3092	0.4407	0.4508	0.4656	0.1555
vit_b16	5	0.5239	0.7792	0.8547	0.3183	0.2342
vit_b16	11	0.5154	0.8022	0.9516	0.3470	0.3069
deit_s16	0	0.2041	0.2774	0.3099	0.3825	0.1377
deit_s16	5	0.4852	0.7375	0.8548	0.3191	0.2635
deit_s16	11	0.4808	0.8488	0.9636	0.3329	0.3692

Frozen-attention 只保留 A dV。因此 Endpoint 与 JVP error 分别测量它在输入缩放方向和 8 个冻结 Rademacher 方向上遗漏的 routing differential。以下检验先在图内按层和 head 聚合，再以图像为统计单位；Holm family 按模型冻结。

模型	Domain	类型	冻结检验	图像级中位数	正/负/平	Holm p	判定
vit_b16	attention	attention-head	late_minus_early_routing_share	0.0523	360/40/0	1.715e-65	SUPPORTED_DIFFERENCE
vit_b16	attention	attention-head	late_minus_early_frozen_attention_endpoint_error	0.1441	383/17/0	5.545e-91	SUPPORTED_DIFFERENCE
vit_b16	attention	attention-head	saturation_routing_relation	-0.2937	5/395/0	2.611e-109	SUPPORTED_DIFFERENCE
vit_b16	attention	attention-head	saturation_error_relation	-0.2238	12/388/0	7.108e-98	SUPPORTED_DIFFERENCE
vit_b16	residual	attention	late_minus_early_identity_share	0.1242	38/362/0	4.068e-67	SUPPORTED_DIFFERENCE
vit_b16	residual	attention	late_minus_early_path_cancellation	-0.0105	173/227/0	7.970e-03	SUPPORTED_DIFFERENCE
vit_b16	residual	mlp	late_minus_early_identity_share	0.1798	396/4/0	1.644e-111	SUPPORTED_DIFFERENCE
vit_b16	residual	mlp	late_minus_early_path_cancellation	-0.0688	304/96/0	2.840e-26	SUPPORTED_DIFFERENCE
deit_s16	attention	attention-head	late_minus_early_routing_share	0.1355	399/1/0	9.317e-118	SUPPORTED_DIFFERENCE
deit_s16	attention	attention-head	late_minus_early_frozen_attention_endpoint_error	0.2896	400/0/0	3.098e-120	SUPPORTED_DIFFERENCE
deit_s16	attention	attention-head	saturation_routing_relation	-0.3143	15/379/6	5.158e-92	SUPPORTED_DIFFERENCE
deit_s16	attention	attention-head	saturation_error_relation	-0.2286	52/338/10	1.794e-52	SUPPORTED_DIFFERENCE
deit_s16	residual	attention	late_minus_early_identity_share	0.0449	48/352/0	6.053e-58	SUPPORTED_DIFFERENCE
deit_s16	residual	attention	late_minus_early_path_cancellation	-0.0074	171/229/0	4.314e-03	SUPPORTED_DIFFERENCE
deit_s16	residual	mlp	late_minus_early_identity_share	0.1780	397/3/0	1.652e-113	SUPPORTED_DIFFERENCE
deit_s16	residual	mlp	late_minus_early_path_cancellation	-0.0588	103/297/0	6.661e-23	SUPPORTED_DIFFERENCE

Saturation 关系是在每张图、每一层的 heads 内计算 Spearman 后再汇总，避免把共同层深趋势当成 head 内关系。其方向与支持状态由上表直接给出，不预设 softmax 越尖锐时 routing omission 越大。

Residual identity/branch

模型	Residual	Block	Identity share	Path cancellation	Identity-branch cosine	Output-identity patch TV
vit_b16	attention	0	0.8847	0.1465	-0.3315	0.0150
vit_b16	attention	5	0.6100	0.2381	0.0606	0.1089
vit_b16	attention	11	0.4442	0.3578	-0.0506	0.1803
vit_b16	mlp	0	0.2225	0.1654	0.1849	0.1398
vit_b16	mlp	5	0.5648	0.2450	0.1321	0.1287
vit_b16	mlp	11	0.6418	0.6333	-0.9071	0.1023
deit_s16	attention	0	0.8729	0.1820	-0.4181	0.0284
deit_s16	attention	5	0.6775	0.1983	0.1826	0.0798
deit_s16	attention	11	0.6559	0.2538	0.0757	0.1047
deit_s16	mlp	0	0.3078	0.2831	-0.0186	0.1798
deit_s16	mlp	5	0.6257	0.1886	0.3342	0.0923
deit_s16	mlp	11	0.6079	0.5878	-0.7599	0.1276

Identity share 描述 absolute Endpoint mass 在 skip 与 branch 两条局部路径间的比例；path cancellation 比较两条 signed map 相加前后的 absolute mass。Output-identity patch TV 衡量 residual 输出相对只走 identity 时的空间依赖

改变。ViT-B/16 中 attention/MLP residual 的最小与最大空间 TV 位于 blocks 0/11 和 8/10；DeiT-S 对应为 1/4 和 9/0。这些层级定位来自冻结 cohort 的描述性中位数。

模型	Phase	Attention−MLP identity share	Attention−MLP path cancellation	Attention−MLP spatial TV
vit_b16	early	0.3382	0.0004	-0.0902
vit_b16	middle	0.0327	0.0165	-0.0050
vit_b16	late	0.0376	-0.0664	-0.0173
deit_s16	early	0.2697	-0.0564	-0.0796
deit_s16	middle	0.0280	0.0009	-0.0026
deit_s16	late	0.0512	-0.0041	-0.0047

模型	Residual	Branch share–path cancellation Spearman (跨 12 blocks)
vit_b16	attention	0.5140
vit_b16	mlp	-0.2657
deit_s16	attention	0.3182
deit_s16	mlp	0.4650

前一表直接比较 attention 与 MLP residual；后一表回答 branch mass 较大时 path cancellation 是否一并变化。两表均为冻结 A1 描述性分析，不加入原 M4 Holm families。

Signed、absolute 与 cancellation

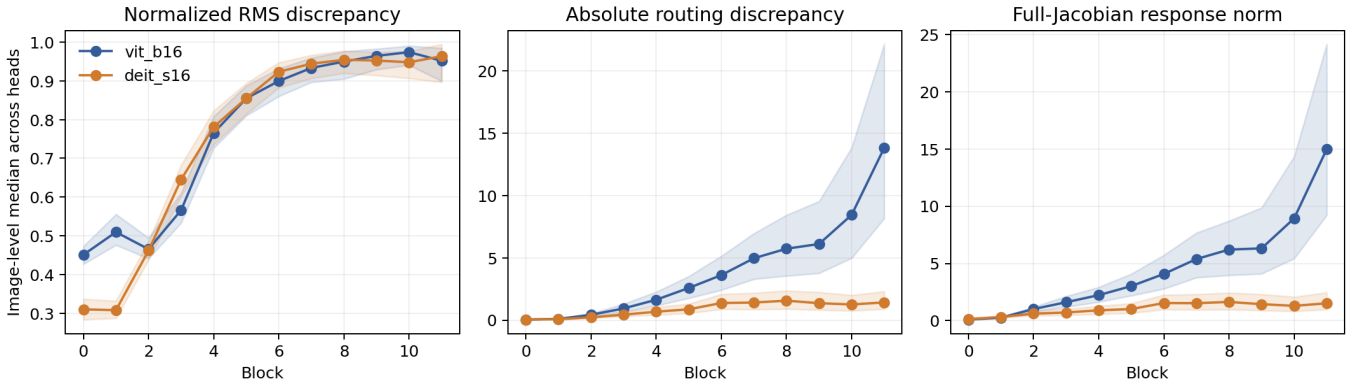
每条路径均同时保留 signed sum、absolute mass、positive/negative mass 和 map cancellation。M4 区分同一 map 内的正负抵消与 routing/value 或 identity/branch 之间的 path cancellation；二者没有合并为单一 “重要性” 指标。所有零分母、零范数和常数 head 关系严格传播为 NA，没有加入 epsilon。

模型	路径	Block	Signed magnitude	Absolute mass	Map cancellation
vit_b16	routing	0	0.0362	2.6596	0.9867
vit_b16	routing	5	0.0600	33.7856	0.9982
vit_b16	routing	11	0.0608	40.7842	0.9982
vit_b16	value	0	0.0306	7.6041	0.9952
vit_b16	value	5	0.0380	28.0522	0.9986
vit_b16	value	11	0.0654	39.0241	0.9981
vit_b16	attention_identity	0	1.3315	495.2271	0.9972
vit_b16	attention_identity	5	0.9816	387.1208	0.9977
vit_b16	attention_identity	11	0.8778	313.3689	0.9974
vit_b16	attention_branch	0	0.3930	63.8344	0.9934
vit_b16	attention_branch	5	0.4140	250.3045	0.9983
vit_b16	attention_branch	11	0.6133	398.0202	0.9986
vit_b16	mlp_identity	0	0.6860	130.4998	0.9945
vit_b16	mlp_identity	5	1.0976	334.0699	0.9971
vit_b16	mlp_identity	11	1.8214	827.8407	0.9977
vit_b16	mlp_branch	0	0.6309	438.0538	0.9986
vit_b16	mlp_branch	5	0.4504	271.9139	0.9983
vit_b16	mlp_branch	11	0.7961	460.2701	0.9981
deit_s16	routing	0	0.0756	4.5294	0.9829
deit_s16	routing	5	0.0506	38.3865	0.9986
deit_s16	routing	11	0.0504	35.9257	0.9986
deit_s16	value	0	0.0818	18.4261	0.9950
deit_s16	value	5	0.0776	40.1955	0.9979
deit_s16	value	11	0.0586	38.8767	0.9985
deit_s16	attention_identity	0	1.6888	459.3077	0.9960
deit_s16	attention_identity	5	1.1087	370.3635	0.9969
deit_s16	attention_identity	11	1.1539	372.6598	0.9966
deit_s16	attention_branch	0	0.4543	63.9204	0.9922

模型	路径	Block	Signed magnitude	Absolute mass	Map cancellation
deit_s16	attention_branch	5	0.4112	169.7186	0.9974
deit_s16	attention_branch	11	0.3639	185.3893	0.9980
deit_s16	mlp_identity	0	0.8741	170.6143	0.9946
deit_s16	mlp_identity	5	1.1847	317.7498	0.9962
deit_s16	mlp_identity	11	1.7620	614.5721	0.9973
deit_s16	mlp_branch	0	0.7619	412.5634	0.9981
deit_s16	mlp_branch	5	0.3466	197.6662	0.9982
deit_s16	mlp_branch	11	0.4532	354.6503	0.9988

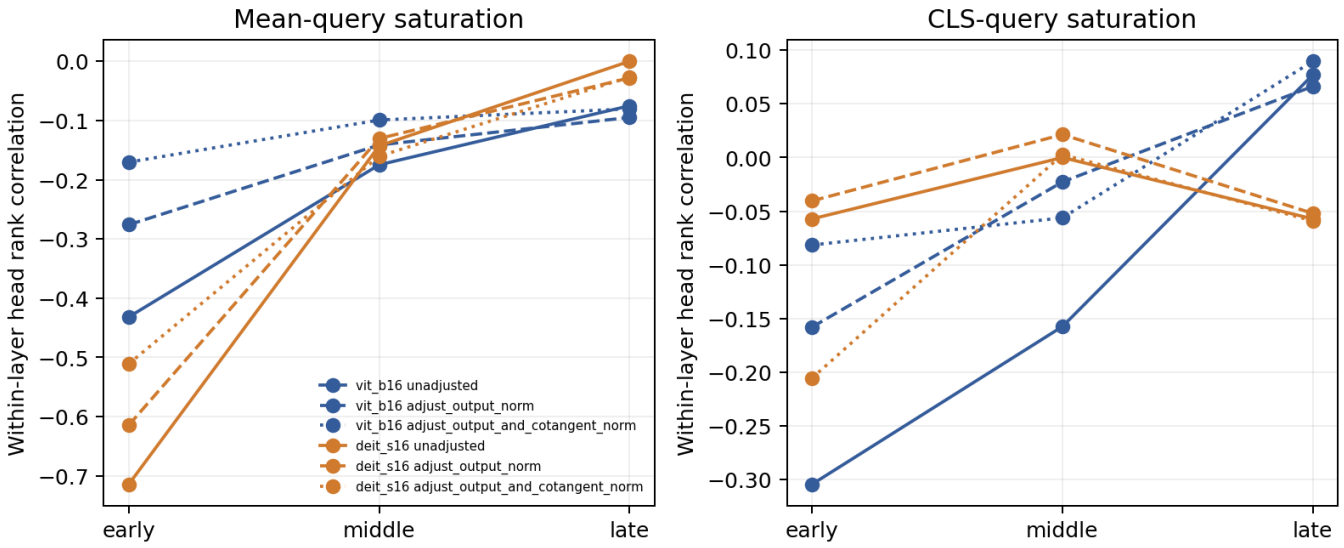
Review-R2 敏感性与 estimand 收紧

Review-R2 检验 late-block Value-only JVP discrepancy 是否由小分母造成。Block 11 的 relative discrepancy 中位数在 ViT-B/16 和 DeiT-S/16 中为 0.9516 和 0.9636；对应的 full-response denominator 为 14.9711 和 1.5155，absolute discrepancy 为 13.8064 和 1.4228。因此，差异与完整响应处于相同 norm 尺度，并非普遍来自接近零的分母。



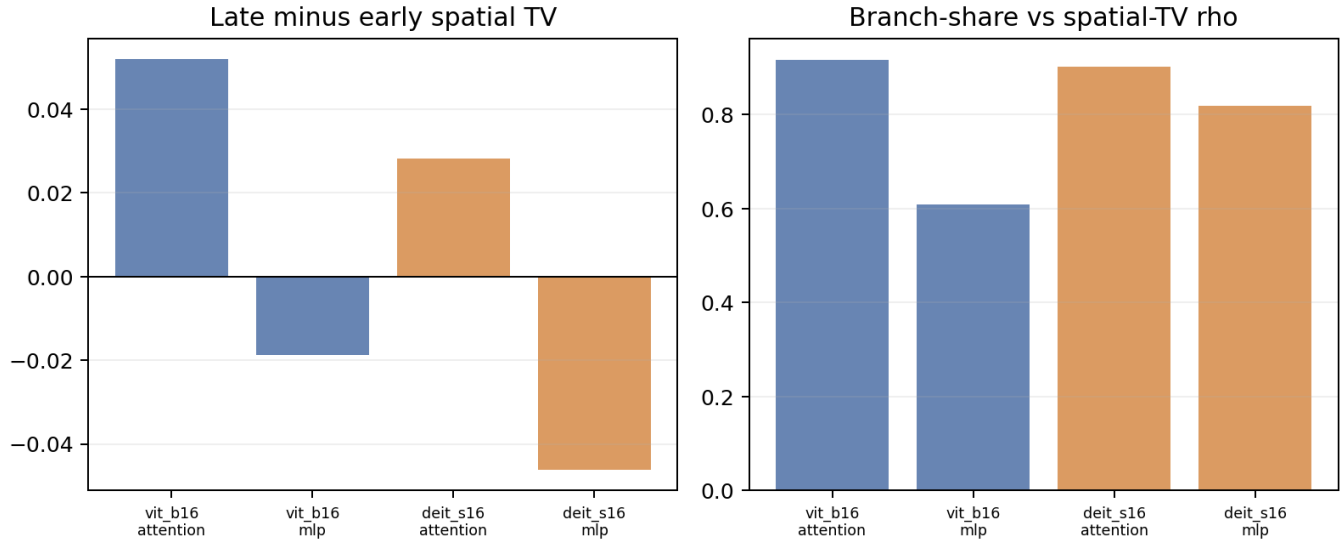
M4 Review Archived R2 Figure 1. 两模型 block 11 的 normalized JVP discrepancy、absolute discrepancy 和 full-response denominator 分布。

控制 head projection-input activation norm 和 downstream-cotangent norm 后，两模型的 mean-query saturation 仍与 routing share、Endpoint error 和 JVP error 呈负相关。ViT-B/16 与 DeiT-S/16 的 fully adjusted correlations 依次为 $-0.1442/-0.3257$、$-0.1129/-0.2185$ 和 $-0.1453/-0.2786$。CLS-query 关系更弱：ViT-B 的 adjusted routing share 和 Endpoint-error 检验均为 Holm $p = 0.0629$。因此正式 summary 使用 all-query mean saturation，并把关联主要定位在 early/middle blocks。



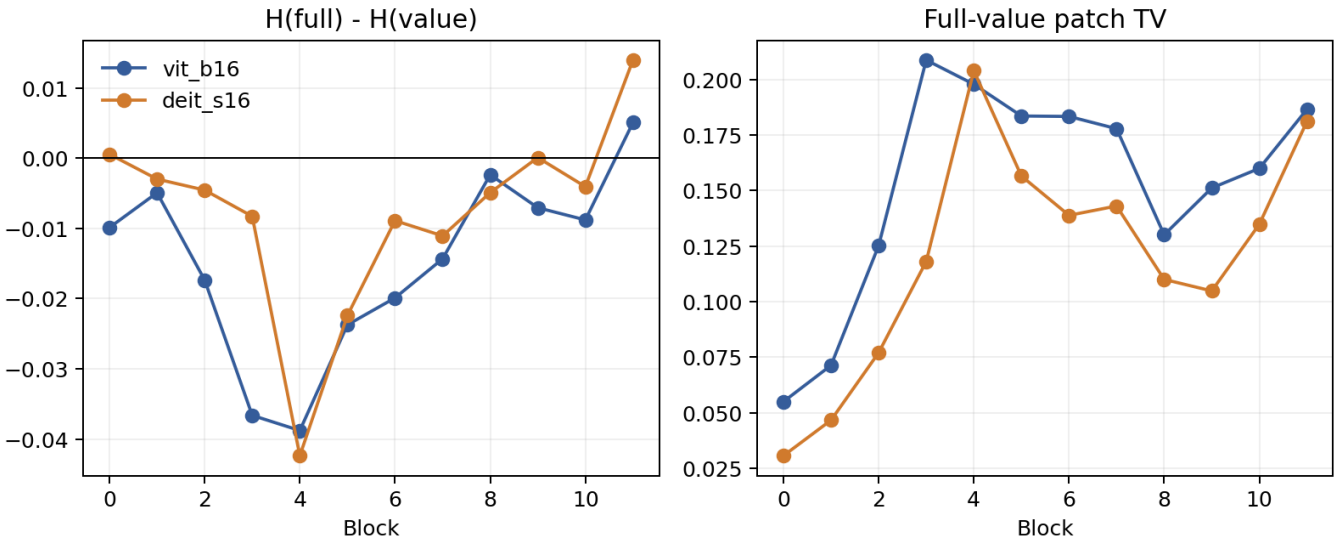
M4 Review Archived R2 Figure 2. Saturation 的 unadjusted/partial-rank 关系、phase sensitivity 与较弱的 CLS-query 结果。

全 cohort 空间分析使用 output 与 identity 的归一化 14×14 patch absolute distributions 之间的 total variation。Attention residual 的 late-minus-early 差在 ViT-B/16 和 DeiT-S/16 中为 +0.0520（[0.0488, 0.0547]）和 +0.0282（[0.0257, 0.0300]）；MLP residual 为 −0.0187（[−0.0220, −0.0150]）和 −0.0462（[−0.0516, −0.0412]）。四项 post-hoc sensitivity tests 全部通过 Holm 校正，branch-relative path scale 与 spatial TV 在图内跨层也呈正相关。



M4 Review Archived R2 Figure 3. Attention 与 MLP 子层的全 cohort residual spatial-change sensitivity。

在每模型 16 张事前保留的 class-logit maps 上，Full–Value patch TV 从 block 0 的 0.0550/0.0307 增至 block 11 的 0.1865/0.1811。其 12-block trajectory 与 M3 entropy、locality 和 distance 方向相容；但 M3 使用 100 图 token-channel targets，且每模型仅一张图重叠。因此这里只形成模型级 trajectory bridge，不构成 routing 导致空间扩展的逐单元或因果联系。



M4 Review Archived R2 Figure 4. M3 空间轨迹与 Full–Value class-logit map 差异之间的描述性桥接。

图

- Image #1（m4_attention_depth）：逐 block 展示 routing share、frozen-attention Endpoint error 与 8-direction JVP error；阴影为 10,000 次图像 bootstrap 95% CI。
- Image #2（m4_attention_saturation）：展示 block-level saturation 与 routing/error 的描述性位置；正式相关检验仍使用图内 layer-wise head Spearman。
- Image #3（m4_residual_depth）：同时展示 attention/MLP residual 的 identity share、path cancellation 和空间 TV。

解释边界

M4 定量描述当前输入点和冻结 ground-truth logit 下的局部 Jacobian 路径记账。较高 routing share 或 branch mass 表示该路径承载较多局部输入依赖；因果重要性仍需独立有限干预证据。

S7. Complete supplementary image index

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- S02 — S02_figure_02_endpoint_cnn_and_path_method_identity
- S03 — S03_figure_03_same_target_endpoint_versus_path
- S04 — S04_figure_04_original_cnn_endpoint_gates
- S05 — S05_figure_05_attention_q_k_v_differential_paths
- S06 — S06_figure_06_residual_edge_accounting
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- S12 — S12_figure_12_sloth_bear_class_logit_maps_across_models
- S13 — S13_figure_13_sloth_bear_multi_target_atlas
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- S15 — S15_figure_15_structural_closure_contracts
- S16 — S16_figure_16_formal_closure_and_negative_controls
- S17 — S17_figure_17_sloth_bear_structural_component_maps
- S18 — S18_figure_18_condition_scaled_analytic_jvp_checks
- S19 — S19_figure_19_matched_finite_activation_replacement
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- S21 — S21_figure_21_rq3_threshold_zoom_and_boundary_exception
- S22 — S22_figure_22_endpoint_and_registered_secondary_predictors
- S23 — S23_figure_23_spatial_intervention_protocol
- S24 — S24_figure_24_sloth_bear_block_5_maps_for_spatial_ranking
- S25 — S25_figure_25_one_image_spatial_effect_curves
- S26 — S26_figure_26_c4b_mandatory_baseline_contrasts
- S27 — S27_figure_27_rq4_family_summary
- S28 — S28_figure_28_identity_correctness_utility_conclusion
- S29 — S29_figure_29_m3_block_trajectories
- S30 — S30_figure_30_m3_cls_minus_patch_entropy_by_block
- S31 — S31_figure_31_m3_cancellation_scales
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- S35 — S35_figure_35_block5_structural_learning_images
- S36 — S36_figure_36_spatial_ranking_learning_images
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- S40 — S40_figure_40_vit_b16_module_chain
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- S84 — S84_supp_m6_04_m6_residual_cases
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S8. 当前定性比较

六项 CPU image-model 计算使用相同来源图像、checkpoint 与预处理。Grad-CAM 使用最后一个 norm1 的 patch activation；attention rollout 与 raw attention 不依赖类别。展示样本不进入总体推断。具体规则见显示协议与计算收

据。

修改意见 v2 将下列展示性或重复性图件从主文移至补充材料；文件名保留原 publication-R2 编号以维持哈希和来源绑定：

- 坐标初始化
- 局部导数传播
- Patch embedding
- DeiT-S/16 十二层图
- ViT-B/16 十二层图
- 模块级学习图像与前向 activation
- 全部 attention heads
- 跨模型展示样本
- DeiT-S/16 方法展示
- ViT-B/16 方法展示
- Positive、signed 与 absolute 读图对照

Table S6. 跨 reference 排名一致性。

Model	Quantity	Spearman	Kendall tau-b	Top-1 agreement	Top-3 overlap	Mean abs rank displacement
DeiT-S/16	finite_effect	0.3071	0.2444	0.4167	0.6111	1.5278
DeiT-S/16	taylor	0.3000	0.2389	0.3333	0.6111	1.5278
ViT-B/16	finite_effect	0.3759	0.2803	0.3333	0.5000	2.9444
ViT-B/16	taylor	0.3782	0.2828	0.3333	0.5000	2.9444

新增补充实验 S9–S14

本补充完成建议文档第十一节的最低实验包与两项 P0 修正。新增结果为补充描述性分析；原确认性决策及封存证据保留。所有实验先冻结 cohort、通道、初始化种子和统计规则，再运行模型。

S9. 随机区域基线修正

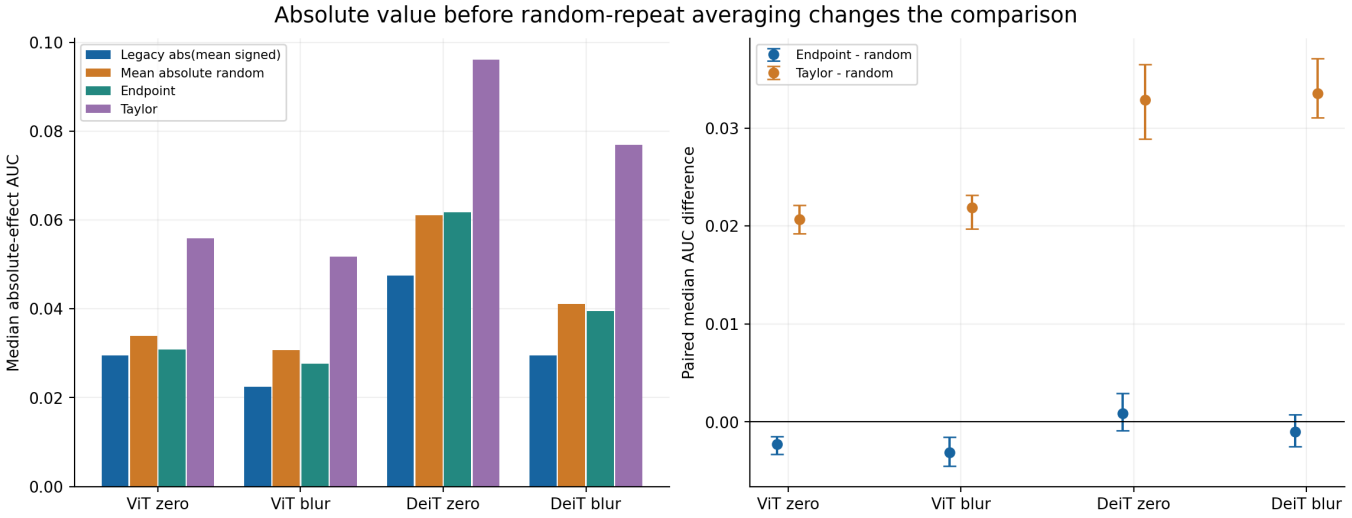
每个随机重复先对效应取绝对值并计算 AUC，再平均 20 次重复及图内 heads。旧估计器先平均 signed curves，因正负抵消低估 absolute-effect random baseline。下表比较 400 图 family 的图像级中位数及配对中位差；区间为 2,000 次图像 bootstrap。

$$\widehat{A}_{\text{random}} = \frac{1}{R}\sum_{r=1}^{R}\frac{1}{0.25}\int_{0}^{0.25}|e_r(a)|\,da. \tag{1}$$

Model	Reference	Legacy random median	Corrected random mean	Corrected random median	Endpoint - random median [95% CI]	Taylor - random median [95% CI]
ViT-B/16	normalized-zero	0.0294	0.0366	0.0339	-0.0023 [-0.0033, -0.0015]	0.0207 [0.0192, 0.0221]
ViT-B/16	blur	0.0225	0.0355	0.0307	-0.0031 [-0.0045, -0.0016]	0.0219 [0.0197, 0.0232]
DeiT-S/16	normalized-zero	0.0475	0.0728	0.0611	0.0009 [-0.0009, 0.0029]	0.0329 [0.0289, 0.0365]

Model	Reference	Legacy random median	Corrected random mean	Corrected random median	Endpoint - random median [95% CI]	Taylor - random median [95% CI]
DeiT-S/16	blur	0.0296	0.0505	0.0411	-0.0010 [-0.0025, 0.0007]	0.0336 [0.0311, 0.0371]

修正后 ViT-B 两种 reference 的 Endpoint−random 中位差均为负；DeiT-S 的两个区间跨零。Taylor−random 四个区间均为正。Endpoint−Taylor 比较保持不变。



每图随机均值的相对 MC 标准误中位数为 5.1%–6.9%，q90 为 7.6%–8.9%。从现有 20 次重复进行 5/10/20 次有放回重采样，误差随重复数减少而增大。固定图像下 cohort mean 的 Monte Carlo 误差小于图像抽样不确定性；单图细小优势仍不适合据此判定。该诊断只覆盖已观察的随机重复，不能证明任意精度下 20 次都足够。

Model	Reference	Median relative MC SE	q90 relative MC SE	MC image error R=5	R=10	R=20
ViT-B/16	normalized-zero	0.0506	0.0759	0.0020	0.0014	0.0010
ViT-B/16	blur	0.0665	0.0883	0.0026	0.0018	0.0013
DeiT-S/16	normalized-zero	0.0622	0.0851	0.0046	0.0032	0.0023
DeiT-S/16	blur	0.0687	0.0895	0.0035	0.0024	0.0017

20-repeat cohort distribution · Per-image repeats · Paired image values, MC SE, split-half and leave-one-repeat-out

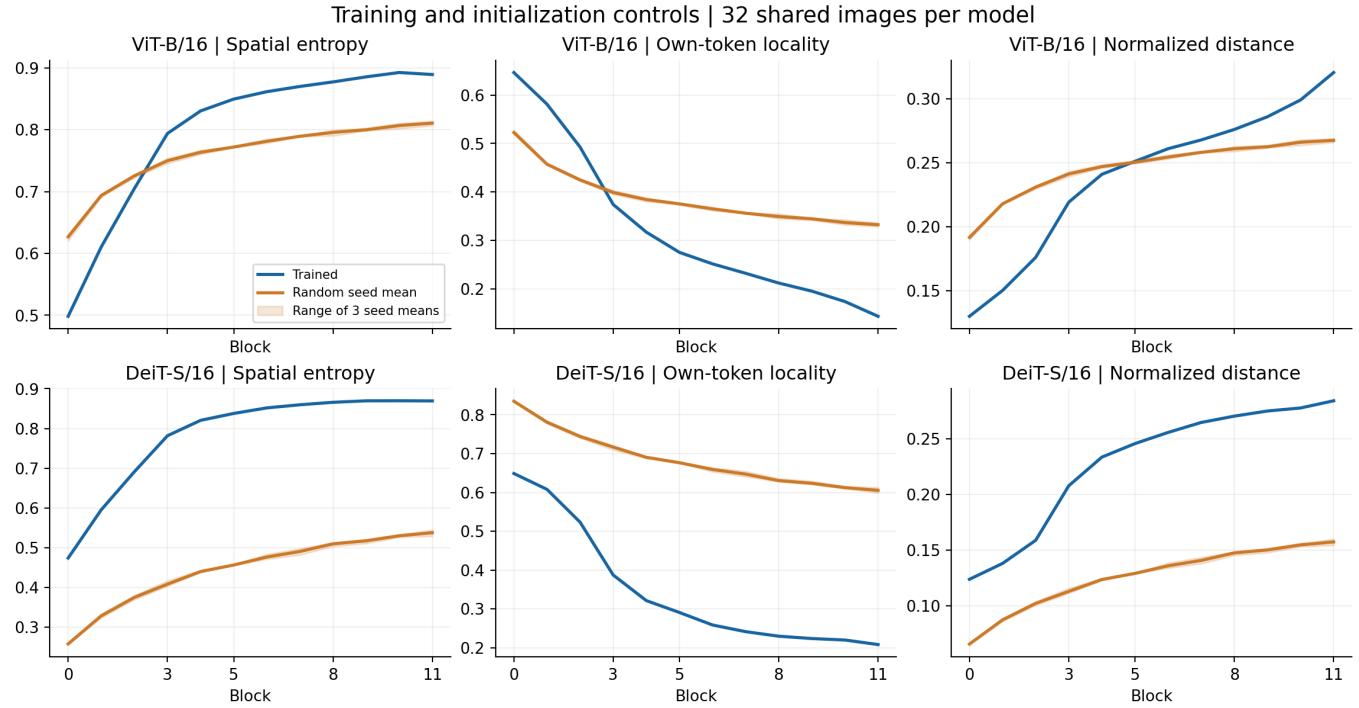
S10. 训练与随机初始化

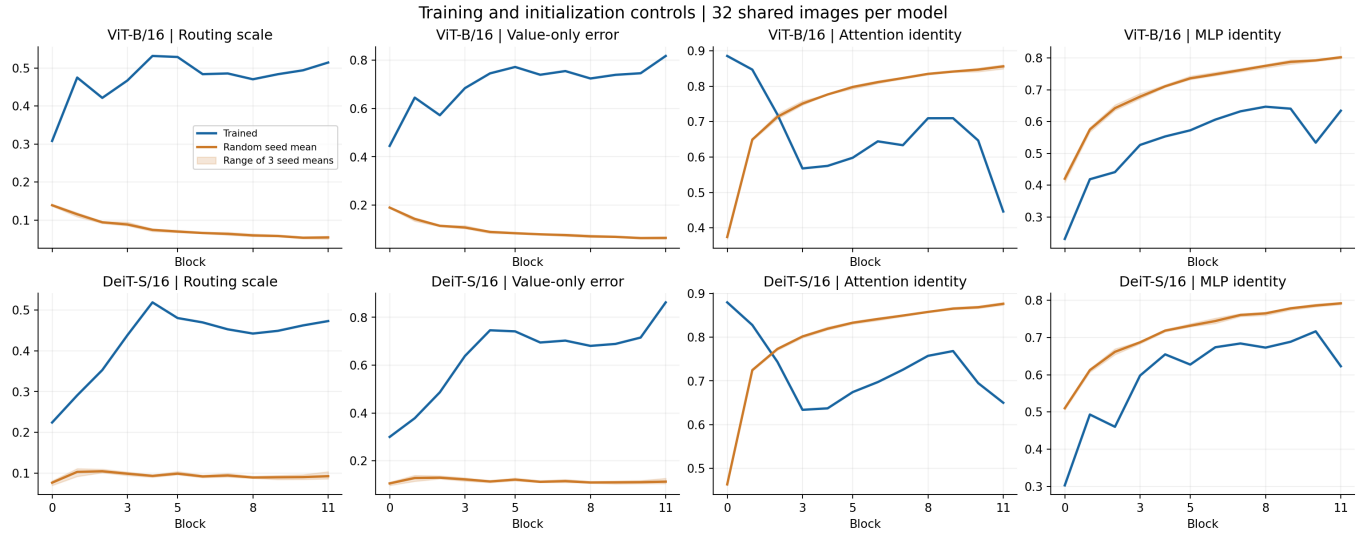
ViT-B/16 和 DeiT-S/16 各使用 3 个随机初始化状态（202609201、202609202、202609203），与训练模型共享原 M3 cohort 的前 32 图、全部 12 blocks、16 tokens 和每层 16 个冻结 channels，M4 覆盖全部 heads。随机模型采用相同 timm 0.9.16 架构与默认初始化；同一状态用于全部图像。训练结果由同图、同目标的原始 M3/M4 数组筛选得到。图内先聚合，early=0–3、late=8–11；训练 − 随机区间同时重采样共享图像和模型种子（2,000 次），保留种子层级。

随机模型已出现 entropy 增大、locality 减小及距离增大，表明空间扩展并非训练参数状态独有。训练模型的 entropy late−early 为 0.2288/0.2256，随机参数状态均值为 0.0940/0.1718；两种参数状态的扩展幅度存在差异。训练参数状态的 routing scale 增大，而随机参数状态均值为负；attention identity 的层深趋势也在两种参数状态间反向。MLP identity 在两类状态中均增加，ViT-B 的训练参数状态减随机参数状态区间包含零。三个随机状态限制了对初始化总体的推断；该对照不识别训练过程的因果效应，也不单独识别架构与初始化的贡献。

Model	Study	Metric	Trained late - early	Random late - early	Trained - random [95% CI]	Random seed 1 / 2 / 3
DeiT-S/16	m3	cancellation	0.0020	0.0036	-0.0016 [-0.0029, -0.0005]	0.0032 / 0.0028 / 0.0050
DeiT-S/16	m3	cls_minus_patch_entropy	-0.1685	-0.1717	-0.0968 [-0.1051, -0.0889]	-0.1697 / -0.1740 / -0.1714
DeiT-S/16	m3	distance	0.1276	0.0576	0.0700 [0.0624, 0.0781]	0.0574 / 0.0584 / 0.0569
DeiT-S/16	m3	entropy	0.2256	0.1718	0.0539 [0.0473, 0.0602]	0.1699 / 0.1740 / 0.1714
DeiT-S/16	m3	locality	-0.3441	-0.1445	-0.1997 [-0.2115, -0.1882]	-0.1433 / -0.1466 / -0.1436
DeiT-S/16	m4	attention_identity_scale	0.0579	0.1188	-0.1767 [-0.1916, -0.1618]	0.1232 / 0.1142 / 0.1190
DeiT-S/16	m4	attention_output_identity_tv	0.0306	-0.0026	0.0331 [0.0274, 0.0388]	-0.0029 / -0.0025 / -0.0023
DeiT-S/16	m4	attention_path_cancellation	0.0958	-0.0042	0.1001 [0.0890, 0.1115]	0.0002 / -0.0036 / -0.0093
DeiT-S/16	m4	attention_residual_path_cancellation	0.0188	-0.0848	0.1036 [0.0810, 0.1271]	-0.0889 / -0.0770 / -0.0884
DeiT-S/16	m4	jvp_discrepancy	0.5465	-0.0944	0.6409 [0.6150, 0.6673]	-0.0871 / -0.0859 / -0.1101
DeiT-S/16	m4	mlp_identity_scale	0.2063	0.1461	0.0602 [0.0308, 0.0907]	0.1423 / 0.1431 / 0.1528
DeiT-S/16	m4	mlp_output_identity_tv	-0.0610	-0.0041	-0.0569 [-0.0722, -0.0413]	-0.0047 / -0.0042 / -0.0034
DeiT-S/16	m4	mlp_residual_path_cancellation	-0.0381	-0.0756	0.0375 [0.0101, 0.0675]	-0.0731 / -0.0761 / -0.0775
DeiT-S/16	m4	routing_scale	0.1314	-0.0043	0.1357 [0.1151, 0.1564]	0.0023 / -0.0032 / -0.0120
DeiT-S/16	m4	saturation_error	0.7536	-0.0348	0.7884 [0.6387, 0.9354]	-0.0348 / -0.0348 / -0.0348
DeiT-S/16	m4	saturation_routing	0.7598	-0.0348	0.7946 [0.6417, 0.9491]	-0.0357 / -0.0339 / -0.0348
DeiT-S/16	m4	value_error	0.2676	-0.0088	0.2764 [0.2406, 0.3122]	-0.0008 / -0.0071 / -0.0184
ViT-B/16	m3	cancellation	0.0024	0.0008	0.0016 [0.0008, 0.0023]	0.0014 / 0.0010 / 0.0000
ViT-B/16	m3	cls_minus_patch_entropy	-0.2741	-0.0940	-0.1802 [-0.1893, -0.1709]	-0.0933 / -0.0970 / -0.0917
ViT-B/16	m3	distance	0.1292	0.0399	0.0893 [0.0833, 0.0953]	0.0399 / 0.0410 / 0.0387
ViT-B/16	m3	entropy	0.2288	0.0940	0.1348 [0.1232, 0.1466]	0.0934 / 0.0968 / 0.0918
ViT-B/16	m3	locality	-0.3522	-0.1005	-0.2517 [-0.2598, -0.2431]	-0.1001 / -0.1036 / -0.0979
ViT-B/16	m4	attention_identity_scale	0.1144	0.1631	-0.2775 [-0.3004, -0.2520]	0.1629 / 0.1662 / 0.1603
ViT-B/16	m4	attention_output_identity_tv	0.0507	-0.0046	0.0553 [0.0454, 0.0644]	-0.0042 / -0.0040 / -0.0058
ViT-B/16	m4	attention_path_cancellation	0.0797	-0.0421	0.1218 [0.1136, 0.1303]	-0.0428 / -0.0413 / -0.0423
ViT-B/16	m4	attention_residual_path_cancellation	0.0138	-0.1050	0.1188 [0.0932, 0.1447]	-0.1028 / -0.1084 / -0.1038
ViT-B/16	m4	jvp_discrepancy	0.4501	-0.2600	0.7101 [0.6739, 0.7465]	-0.2621 / -0.2548 / -0.2631
ViT-B/16	m4	mlp_identity_scale	0.1951	0.1819	0.0132 [-0.0093, 0.0361]	0.1829 / 0.1851 / 0.1775
ViT-B/16	m4	mlp_output_identity_tv	-0.0205	-0.0042	-0.0163 [-0.0295, -0.0047]	-0.0041 / -0.0045 / -0.0041
ViT-B/16	m4	mlp_residual_path_cancellation	-0.0887	-0.0866	0.1753 [0.1350, 0.2174]	-0.0845 / -0.0907 / -0.0846
ViT-B/16	m4	routing_scale	0.0512	-0.0479	0.0991 [0.0810, 0.1160]	-0.0485 / -0.0470 / -0.0481
ViT-B/16	m4	saturation_error	0.3568	0.0232	0.3336 [0.2370, 0.4496]	0.0827 / -0.0498 / 0.0366

Model	Study	Metric	Trained late - early	Random late - early	Trained - random [95% CI]	Random seed 1 / 2 / 3
ViT-B/16	m4	saturation_routing	0.4379	0.0203	0.4177 [0.3136, 0.5211]	0.0636 / -0.0431 / 0.0402
ViT-B/16	m4	value_error	0.1437	-0.0623	0.2061 [0.1689, 0.2420]	-0.0621 / -0.0615 / -0.0633





Early/late and difference estimates · Image values

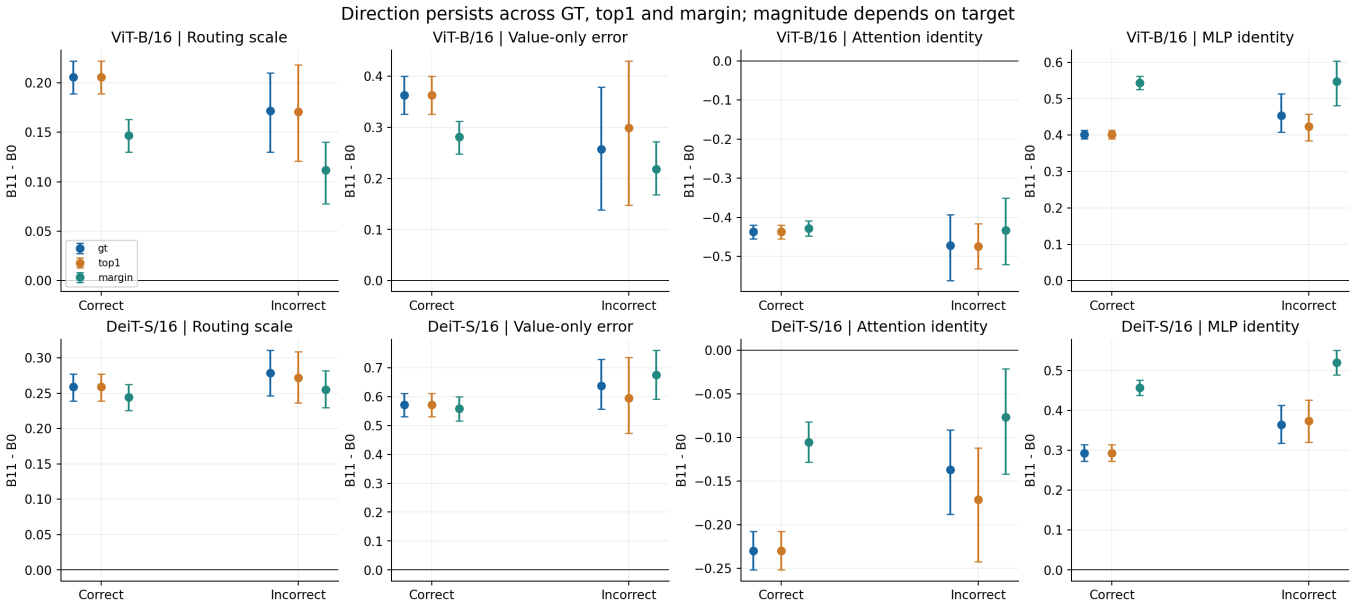
S11. 目标稳健性

原 M3 冻结 100 图/模型，GT、clean-top1、top1−runnerup 三目标，blocks 0/5/11、全部 heads。分类索引固定于 clean 输入。正确/错误图分别为 ViT-B 92/8、DeiT-S 86/14；正确图的 GT 与 top1 完全一致。下表报告每图 head 聚

合后的 B11−B0，并按 all/correct/incorrect 分组；所有目标在同一组使用相同 bootstrap 图像索引。

routing scale 和 Value-only error 在三种目标中均随深度增加；attention identity 减少，MLP identity 增加，正确与错误子集方向一致。Margin 改变幅度，例如 ViT-B routing 增量由 GT 的 0.2031 降至 0.1439，DeiT-S attention identity 降幅由 −0.2168 变为 −0.1014。两模型错误子集只有 8/14 图，其区间相应反映较少的图像信息。

native attention-output JVP 只依赖模型与输入，不依赖被解释的类别；其 8 方向 discrepancy 从原 M4 同图记录继承。target_contracted_jvp 另用每个目标对 head 输出的 cotangent 收缩方向导数，衡量目标相关 Value-only discrepancy。两者作为不同指标报告。Saturation relation 为图内同 block 跨 head 的 Spearman，并报告其层深变化；完整 block 值、目标配对差和 NA 数量见数据文件。



Model	Subset	Target	Metric	n	B11 - B0 mean	95% CI
ViT-B/16	all	gt	routing__scale	100	0.2031	[0.1868, 0.2189]
ViT-B/16	all	top1	routing__scale	100	0.2030	[0.1862, 0.2184]
ViT-B/16	all	margin	routing__scale	100	0.1439	[0.1283, 0.1595]
ViT-B/16	all	gt	value__error	100	0.3539	[0.3170, 0.3877]
ViT-B/16	all	top1	value__error	100	0.3572	[0.3203, 0.3917]
ViT-B/16	all	margin	value__error	100	0.2762	[0.2448, 0.3059]
ViT-B/16	all	gt	jvp__discrepancy	100	0.4757	[0.4569, 0.4942]
ViT-B/16	all	top1	jvp__discrepancy	100	0.4757	[0.4569, 0.4942]
ViT-B/16	all	margin	jvp__discrepancy	100	0.4757	[0.4569, 0.4942]
ViT-B/16	all	gt	attention__path__cancellation	100	0.1587	[0.1469, 0.1699]
ViT-B/16	all	top1	attention__path__cancellation	100	0.1601	[0.1488, 0.1713]
ViT-B/16	all	margin	attention__path__cancellation	100	0.1427	[0.1326, 0.1529]
ViT-B/16	all	gt	saturation__routing	100	-0.1076	[-0.1852, -0.0322]
ViT-B/16	all	top1	saturation__routing	100	-0.1087	[-0.1849, -0.0364]
ViT-B/16	all	margin	saturation__routing	100	0.0102	[-0.0637, 0.0853]
ViT-B/16	all	gt	saturation__error	100	-0.0495	[-0.1332, 0.0347]
ViT-B/16	all	top1	saturation__error	100	-0.0517	[-0.1335, 0.0312]
ViT-B/16	all	margin	saturation__error	100	0.0421	[-0.0353, 0.1223]
ViT-B/16	all	gt	target_contracted_jvp	100	0.3382	[0.3029, 0.3751]
ViT-B/16	all	top1	target_contracted_jvp	100	0.3414	[0.3046, 0.3786]
ViT-B/16	all	margin	target_contracted_jvp	100	0.2887	[0.2568, 0.3201]
ViT-B/16	all	gt	attention__identity_scale	100	-0.4390	[-0.4577, -0.4213]
ViT-B/16	all	top1	attention__identity_scale	100	-0.4392	[-0.4568, -0.4219]
ViT-B/16	all	margin	attention__identity_scale	100	-0.4285	[-0.4479, -0.4085]
ViT-B/16	all	gt	attention__output__identity_tv	100	0.1823	[0.1692, 0.1956]
ViT-B/16	all	top1	attention__output__identity_tv	100	0.1848	[0.1723, 0.1978]
ViT-B/16	all	margin	attention__output__identity_tv	100	0.1233	[0.1128, 0.1340]

Model	Subset	Target	Metric	n	B11 - B0 mean	95% CI
ViT-B/16	all	gt	attention_residual_path_cancellation	100	0.2017	[0.1688, 0.2320]
ViT-B/16	all	top1	attention_residual_path_cancellation	100	0.2042	[0.1719, 0.2345]
ViT-B/16	all	margin	attention_residual_path_cancellation	100	0.0329	[0.0108, 0.0553]
ViT-B/16	all	gt	mlp_identity_scale	100	0.4057	[0.3937, 0.4183]
ViT-B/16	all	top1	mlp_identity_scale	100	0.4033	[0.3919, 0.4144]
ViT-B/16	all	margin	mlp_identity_scale	100	0.5437	[0.5254, 0.5627]
ViT-B/16	all	gt	mlp_output_identity_tv	100	-0.0275	[-0.0369, -0.0182]
ViT-B/16	all	top1	mlp_output_identity_tv	100	-0.0277	[-0.0370, -0.0186]
ViT-B/16	all	margin	mlp_output_identity_tv	100	-0.0897	[-0.0985, -0.0805]
ViT-B/16	all	gt	mlp_residual_path_cancellation	100	0.4451	[0.4210, 0.4662]
ViT-B/16	all	top1	mlp_residual_path_cancellation	100	0.4546	[0.4343, 0.4731]
ViT-B/16	all	margin	mlp_residual_path_cancellation	100	0.1074	[0.0777, 0.1369]
ViT-B/16	correct	gt	routing_scale	92	0.2058	[0.1891, 0.2222]
ViT-B/16	correct	top1	routing_scale	92	0.2058	[0.1891, 0.2222]
ViT-B/16	correct	margin	routing_scale	92	0.1467	[0.1299, 0.1633]
ViT-B/16	correct	gt	value_error	92	0.3623	[0.3256, 0.4001]
ViT-B/16	correct	top1	value_error	92	0.3623	[0.3256, 0.4001]
ViT-B/16	correct	margin	value_error	92	0.2813	[0.2476, 0.3116]
ViT-B/16	correct	gt	jvp_discrepancy	92	0.4829	[0.4657, 0.4998]
ViT-B/16	correct	top1	jvp_discrepancy	92	0.4829	[0.4657, 0.4998]
ViT-B/16	correct	margin	jvp_discrepancy	92	0.4829	[0.4657, 0.4998]
ViT-B/16	correct	gt	attention_path_cancellation	92	0.1605	[0.1480, 0.1723]
ViT-B/16	correct	top1	attention_path_cancellation	92	0.1605	[0.1480, 0.1723]
ViT-B/16	correct	margin	attention_path_cancellation	92	0.1453	[0.1339, 0.1561]
ViT-B/16	correct	gt	saturation_routing	92	-0.1206	[-0.2004, -0.0436]
ViT-B/16	correct	top1	saturation_routing	92	-0.1206	[-0.2004, -0.0436]
ViT-B/16	correct	margin	saturation_routing	92	-0.0131	[-0.0883, 0.0598]
ViT-B/16	correct	gt	saturation_error	92	-0.0628	[-0.1478, 0.0175]
ViT-B/16	correct	top1	saturation_error	92	-0.0628	[-0.1478, 0.0175]
ViT-B/16	correct	margin	saturation_error	92	0.0260	[-0.0562, 0.1081]
ViT-B/16	correct	gt	target_contracted_jvp	92	0.3431	[0.3039, 0.3823]
ViT-B/16	correct	top1	target_contracted_jvp	92	0.3431	[0.3039, 0.3823]
ViT-B/16	correct	margin	target_contracted_jvp	92	0.2953	[0.2618, 0.3283]
ViT-B/16	correct	gt	attention_identity_scale	92	-0.4362	[-0.4546, -0.4194]
ViT-B/16	correct	top1	attention_identity_scale	92	-0.4362	[-0.4546, -0.4194]
ViT-B/16	correct	margin	attention_identity_scale	92	-0.4281	[-0.4478, -0.4091]
ViT-B/16	correct	gt	attention_output_identity_tv	92	0.1884	[0.1759, 0.2016]
ViT-B/16	correct	top1	attention_output_identity_tv	92	0.1884	[0.1759, 0.2016]
ViT-B/16	correct	margin	attention_output_identity_tv	92	0.1259	[0.1143, 0.1374]
ViT-B/16	correct	gt	attention_residual_path_cancellation	92	0.2164	[0.1837, 0.2492]
ViT-B/16	correct	top1	attention_residual_path_cancellation	92	0.2164	[0.1837, 0.2492]
ViT-B/16	correct	margin	attention_residual_path_cancellation	92	0.0374	[0.0136, 0.0611]
ViT-B/16	correct	gt	mlp_identity_scale	92	0.4015	[0.3894, 0.4136]
ViT-B/16	correct	top1	mlp_identity_scale	92	0.4015	[0.3894, 0.4136]
ViT-B/16	correct	margin	mlp_identity_scale	92	0.5434	[0.5249, 0.5620]
ViT-B/16	correct	gt	mlp_output_identity_tv	92	-0.0285	[-0.0381, -0.0189]
ViT-B/16	correct	top1	mlp_output_identity_tv	92	-0.0285	[-0.0381, -0.0189]
ViT-B/16	correct	margin	mlp_output_identity_tv	92	-0.0900	[-0.0996, -0.0807]
ViT-B/16	correct	gt	mlp_residual_path_cancellation	92	0.4577	[0.4381, 0.4765]
ViT-B/16	correct	top1	mlp_residual_path_cancellation	92	0.4577	[0.4381, 0.4765]
ViT-B/16	correct	margin	mlp_residual_path_cancellation	92	0.1127	[0.0840, 0.1427]
ViT-B/16	incorrect	gt	routing_scale	8	0.1719	[0.1298, 0.2101]
ViT-B/16	incorrect	top1	routing_scale	8	0.1706	[0.1210, 0.2183]
ViT-B/16	incorrect	margin	routing_scale	8	0.1120	[0.0779, 0.1399]
ViT-B/16	incorrect	gt	value_error	8	0.2568	[0.1380, 0.3788]
ViT-B/16	incorrect	top1	value_error	8	0.2985	[0.1477, 0.4298]
ViT-B/16	incorrect	margin	value_error	8	0.2183	[0.1684, 0.2715]
ViT-B/16	incorrect	gt	jvp_discrepancy	8	0.3924	[0.2737, 0.5011]
ViT-B/16	incorrect	top1	jvp_discrepancy	8	0.3924	[0.2737, 0.5011]
ViT-B/16	incorrect	margin	jvp_discrepancy	8	0.3924	[0.2737, 0.5011]
ViT-B/16	incorrect	gt	attention_path_cancellation	8	0.1385	[0.1052, 0.1729]
ViT-B/16	incorrect	top1	attention_path_cancellation	8	0.1563	[0.1272, 0.1850]
ViT-B/16	incorrect	margin	attention_path_cancellation	8	0.1125	[0.0926, 0.1367]
ViT-B/16	incorrect	gt	saturation_routing	8	0.0411	[-0.2710, 0.3628]
ViT-B/16	incorrect	top1	saturation_routing	8	0.0271	[-0.2099, 0.2570]

Model	Subset	Target	Metric	n	B11 - B0 mean	95% CI
ViT-B/16	incorrect	margin	saturation_routing	8	0.2780	[-0.0166, 0.5656]
ViT-B/16	incorrect	gt	saturation_error	8	0.1031	[-0.2203, 0.4493]
ViT-B/16	incorrect	top1	saturation_error	8	0.0760	[-0.1198, 0.2850]
ViT-B/16	incorrect	margin	saturation_error	8	0.2273	[0.0402, 0.4187]
ViT-B/16	incorrect	gt	target_contracted_jvp	8	0.2814	[0.1617, 0.3988]
ViT-B/16	incorrect	top1	target_contracted_jvp	8	0.3210	[0.1766, 0.4598]
ViT-B/16	incorrect	margin	target_contracted_jvp	8	0.2126	[0.1223, 0.3130]
ViT-B/16	incorrect	gt	attention_identity_scale	8	-0.4712	[-0.5617, -0.3925]
ViT-B/16	incorrect	top1	attention_identity_scale	8	-0.4740	[-0.5316, -0.4160]
ViT-B/16	incorrect	margin	attention_identity_scale	8	-0.4324	[-0.5205, -0.3503]
ViT-B/16	incorrect	gt	attention_output_identity_tv	8	0.1119	[0.0867, 0.1380]
ViT-B/16	incorrect	top1	attention_output_identity_tv	8	0.1432	[0.1076, 0.1813]
ViT-B/16	incorrect	margin	attention_output_identity_tv	8	0.0931	[0.0694, 0.1182]
ViT-B/16	incorrect	gt	attention_residual_path_cancellation	8	0.0321	[-0.0432, 0.1290]
ViT-B/16	incorrect	top1	attention_residual_path_cancellation	8	0.0640	[-0.0001, 0.1464]
ViT-B/16	incorrect	margin	attention_residual_path_cancellation	8	-0.0194	[-0.0629, 0.0265]
ViT-B/16	incorrect	gt	mlp_identity_scale	8	0.4537	[0.4078, 0.5139]
ViT-B/16	incorrect	top1	mlp_identity_scale	8	0.4232	[0.3842, 0.4573]
ViT-B/16	incorrect	margin	mlp_identity_scale	8	0.5469	[0.4811, 0.6039]
ViT-B/16	incorrect	gt	mlp_output_identity_tv	8	-0.0169	[-0.0455, 0.0156]
ViT-B/16	incorrect	top1	mlp_output_identity_tv	8	-0.0190	[-0.0503, 0.0132]
ViT-B/16	incorrect	margin	mlp_output_identity_tv	8	-0.0862	[-0.1020, -0.0693]
ViT-B/16	incorrect	gt	mlp_residual_path_cancellation	8	0.3002	[0.1584, 0.4207]
ViT-B/16	incorrect	top1	mlp_residual_path_cancellation	8	0.4195	[0.3318, 0.5030]
ViT-B/16	incorrect	margin	mlp_residual_path_cancellation	8	0.0468	[-0.0459, 0.1747]
DeiT-S/16	all	gt	routing_scale	100	0.2617	[0.2450, 0.2785]
DeiT-S/16	all	top1	routing_scale	100	0.2607	[0.2438, 0.2778]
DeiT-S/16	all	margin	routing_scale	100	0.2460	[0.2302, 0.2625]
DeiT-S/16	all	gt	value_error	100	0.5806	[0.5452, 0.6180]
DeiT-S/16	all	top1	value_error	100	0.5745	[0.5354, 0.6131]
DeiT-S/16	all	margin	value_error	100	0.5744	[0.5352, 0.6135]
DeiT-S/16	all	gt	jvp_discrepancy	100	0.6374	[0.6193, 0.6563]
DeiT-S/16	all	top1	jvp_discrepancy	100	0.6374	[0.6193, 0.6563]
DeiT-S/16	all	margin	jvp_discrepancy	100	0.6374	[0.6193, 0.6563]
DeiT-S/16	all	gt	attention_path_cancellation	100	0.2407	[0.2279, 0.2540]
DeiT-S/16	all	top1	attention_path_cancellation	100	0.2392	[0.2263, 0.2529]
DeiT-S/16	all	margin	attention_path_cancellation	100	0.2318	[0.2209, 0.2430]
DeiT-S/16	all	gt	saturation_routing	100	-0.3766	[-0.4594, -0.2909]
DeiT-S/16	all	top1	saturation_routing	100	-0.3789	[-0.4634, -0.2920]
DeiT-S/16	all	margin	saturation_routing	100	-0.2943	[-0.3863, -0.2006]
DeiT-S/16	all	gt	saturation_error	100	-0.3371	[-0.4417, -0.2337]
DeiT-S/16	all	top1	saturation_error	100	-0.3486	[-0.4509, -0.2480]
DeiT-S/16	all	margin	saturation_error	100	-0.0817	[-0.1794, 0.0137]
DeiT-S/16	all	gt	target_contracted_jvp	100	0.5393	[0.4936, 0.5852]
DeiT-S/16	all	top1	target_contracted_jvp	100	0.5523	[0.5020, 0.6041]
DeiT-S/16	all	margin	target_contracted_jvp	100	0.5375	[0.4887, 0.5895]
DeiT-S/16	all	gt	attention_identity_scale	100	-0.2168	[-0.2369, -0.1960]
DeiT-S/16	all	top1	attention_identity_scale	100	-0.2216	[-0.2418, -0.2006]
DeiT-S/16	all	margin	attention_identity_scale	100	-0.1014	[-0.1224, -0.0796]
DeiT-S/16	all	gt	attention_output_identity_tv	100	0.0821	[0.0723, 0.0918]
DeiT-S/16	all	top1	attention_output_identity_tv	100	0.0859	[0.0759, 0.0961]
DeiT-S/16	all	margin	attention_output_identity_tv	100	0.0337	[0.0255, 0.0420]
DeiT-S/16	all	gt	attention_residual_path_cancellation	100	0.0709	[0.0464, 0.0949]
DeiT-S/16	all	top1	attention_residual_path_cancellation	100	0.0706	[0.0456, 0.0950]
DeiT-S/16	all	margin	attention_residual_path_cancellation	100	-0.0152	[-0.0397, 0.0084]
DeiT-S/16	all	gt	mlp_identity_scale	100	0.3029	[0.2846, 0.3227]
DeiT-S/16	all	top1	mlp_identity_scale	100	0.3041	[0.2847, 0.3235]
DeiT-S/16	all	margin	mlp_identity_scale	100	0.4663	[0.4484, 0.4830]
DeiT-S/16	all	gt	mlp_output_identity_tv	100	-0.0414	[-0.0564, -0.0263]
DeiT-S/16	all	top1	mlp_output_identity_tv	100	-0.0413	[-0.0566, -0.0252]
DeiT-S/16	all	margin	mlp_output_identity_tv	100	-0.1197	[-0.1312, -0.1073]
DeiT-S/16	all	gt	mlp_residual_path_cancellation	100	0.2912	[0.2535, 0.3278]
DeiT-S/16	all	top1	mlp_residual_path_cancellation	100	0.3101	[0.2762, 0.3429]
DeiT-S/16	all	margin	mlp_residual_path_cancellation	100	-0.0462	[-0.0768, -0.0155]
DeiT-S/16	correct	gt	routing_scale	86	0.2590	[0.2389, 0.2772]

Model	Subset	Target	Metric	n	B11 - B0 mean	95% CI
DeiT-S/16	correct	top1	routing_scale	86	0.2590	[0.2389, 0.2772]
DeiT-S/16	correct	margin	routing_scale	86	0.2445	[0.2258, 0.2624]
DeiT-S/16	correct	gt	value_error	86	0.5713	[0.5310, 0.6118]
DeiT-S/16	correct	top1	value_error	86	0.5713	[0.5310, 0.6118]
DeiT-S/16	correct	margin	value_error	86	0.5580	[0.5160, 0.5994]
DeiT-S/16	correct	gt	jvp_discrepancy	86	0.6362	[0.6153, 0.6562]
DeiT-S/16	correct	top1	jvp_discrepancy	86	0.6362	[0.6153, 0.6562]
DeiT-S/16	correct	margin	jvp_discrepancy	86	0.6362	[0.6153, 0.6562]
DeiT-S/16	correct	gt	attention_path_cancellation	86	0.2396	[0.2250, 0.2535]
DeiT-S/16	correct	top1	attention_path_cancellation	86	0.2396	[0.2250, 0.2535]
DeiT-S/16	correct	margin	attention_path_cancellation	86	0.2285	[0.2155, 0.2406]
DeiT-S/16	correct	gt	saturation_routing	86	-0.4027	[-0.4930, -0.3083]
DeiT-S/16	correct	top1	saturation_routing	86	-0.4027	[-0.4930, -0.3083]
DeiT-S/16	correct	margin	saturation_routing	86	-0.3116	[-0.4140, -0.2066]
DeiT-S/16	correct	gt	saturation_error	86	-0.3841	[-0.4930, -0.2638]
DeiT-S/16	correct	top1	saturation_error	86	-0.3841	[-0.4930, -0.2638]
DeiT-S/16	correct	margin	saturation_error	86	-0.0850	[-0.1894, 0.0159]
DeiT-S/16	correct	gt	target_contracted_jvp	86	0.5355	[0.4808, 0.5896]
DeiT-S/16	correct	top1	target_contracted_jvp	86	0.5355	[0.4808, 0.5896]
DeiT-S/16	correct	margin	target_contracted_jvp	86	0.5281	[0.4708, 0.5831]
DeiT-S/16	correct	gt	attention_identity_scale	86	-0.2298	[-0.2518, -0.2078]
DeiT-S/16	correct	top1	attention_identity_scale	86	-0.2298	[-0.2518, -0.2078]
DeiT-S/16	correct	margin	attention_identity_scale	86	-0.1054	[-0.1286, -0.0821]
DeiT-S/16	correct	gt	attention_output_identity_tv	86	0.0885	[0.0775, 0.0999]
DeiT-S/16	correct	top1	attention_output_identity_tv	86	0.0885	[0.0775, 0.0999]
DeiT-S/16	correct	margin	attention_output_identity_tv	86	0.0356	[0.0266, 0.0449]
DeiT-S/16	correct	gt	attention_residual_path_cancellation	86	0.0827	[0.0547, 0.1116]
DeiT-S/16	correct	top1	attention_residual_path_cancellation	86	0.0827	[0.0547, 0.1116]
DeiT-S/16	correct	margin	attention_residual_path_cancellation	86	-0.0110	[-0.0399, 0.0195]
DeiT-S/16	correct	gt	mlp_identity_scale	86	0.2928	[0.2735, 0.3141]
DeiT-S/16	correct	top1	mlp_identity_scale	86	0.2928	[0.2735, 0.3141]
DeiT-S/16	correct	margin	mlp_identity_scale	86	0.4574	[0.4379, 0.4762]
DeiT-S/16	correct	gt	mlp_output_identity_tv	86	-0.0346	[-0.0509, -0.0184]
DeiT-S/16	correct	top1	mlp_output_identity_tv	86	-0.0346	[-0.0509, -0.0184]
DeiT-S/16	correct	margin	mlp_output_identity_tv	86	-0.1145	[-0.1268, -0.1011]
DeiT-S/16	correct	gt	mlp_residual_path_cancellation	86	0.3265	[0.2927, 0.3609]
DeiT-S/16	correct	top1	mlp_residual_path_cancellation	86	0.3265	[0.2927, 0.3609]
DeiT-S/16	correct	margin	mlp_residual_path_cancellation	86	-0.0345	[-0.0692, 0.0001]
DeiT-S/16	incorrect	gt	routing_scale	14	0.2785	[0.2462, 0.3109]
DeiT-S/16	incorrect	top1	routing_scale	14	0.2717	[0.2366, 0.3090]
DeiT-S/16	incorrect	margin	routing_scale	14	0.2551	[0.2297, 0.2818]
DeiT-S/16	incorrect	gt	value_error	14	0.6374	[0.5562, 0.7293]
DeiT-S/16	incorrect	top1	value_error	14	0.5942	[0.4738, 0.7350]
DeiT-S/16	incorrect	margin	value_error	14	0.6752	[0.5912, 0.7607]
DeiT-S/16	incorrect	gt	jvp_discrepancy	14	0.6446	[0.6040, 0.6821]
DeiT-S/16	incorrect	top1	jvp_discrepancy	14	0.6446	[0.6040, 0.6821]
DeiT-S/16	incorrect	margin	jvp_discrepancy	14	0.6446	[0.6040, 0.6821]
DeiT-S/16	incorrect	gt	attention_path_cancellation	14	0.2471	[0.2123, 0.2797]
DeiT-S/16	incorrect	top1	attention_path_cancellation	14	0.2369	[0.1957, 0.2831]
DeiT-S/16	incorrect	margin	attention_path_cancellation	14	0.2521	[0.2257, 0.2803]
DeiT-S/16	incorrect	gt	saturation_routing	14	-0.2163	[-0.4408, 0.0163]
DeiT-S/16	incorrect	top1	saturation_routing	14	-0.2327	[-0.4122, -0.0448]
DeiT-S/16	incorrect	margin	saturation_routing	14	-0.1878	[-0.3878, 0.0286]
DeiT-S/16	incorrect	gt	saturation_error	14	-0.0490	[-0.3469, 0.2367]
DeiT-S/16	incorrect	top1	saturation_error	14	-0.1306	[-0.3633, 0.0939]
DeiT-S/16	incorrect	margin	saturation_error	14	-0.0612	[-0.3674, 0.2327]
DeiT-S/16	incorrect	gt	target_contracted_jvp	14	0.5631	[0.5059, 0.6286]
DeiT-S/16	incorrect	top1	target_contracted_jvp	14	0.6555	[0.4921, 0.8523]
DeiT-S/16	incorrect	margin	target_contracted_jvp	14	0.5956	[0.4935, 0.6992]
DeiT-S/16	incorrect	gt	attention_identity_scale	14	-0.1368	[-0.1879, -0.0917]
DeiT-S/16	incorrect	top1	attention_identity_scale	14	-0.1713	[-0.2423, -0.1122]
DeiT-S/16	incorrect	margin	attention_identity_scale	14	-0.0768	[-0.1418, -0.0212]
DeiT-S/16	incorrect	gt	attention_output_identity_tv	14	0.0426	[0.0273, 0.0589]
DeiT-S/16	incorrect	top1	attention_output_identity_tv	14	0.0700	[0.0394, 0.1080]
DeiT-S/16	incorrect	margin	attention_output_identity_tv	14	0.0219	[0.0026, 0.0449]

Model	Subset	Target	Metric	n	B11 - B0 mean	95% CI
DeiT-S/16	incorrect	gt	attention_residual_path_cancellation	14	-0.0021	[-0.0379, 0.0375]
DeiT-S/16	incorrect	top1	attention_residual_path_cancellation	14	-0.0041	[-0.0474, 0.0455]
DeiT-S/16	incorrect	margin	attention_residual_path_cancellation	14	-0.0407	[-0.0834, 0.0012]
DeiT-S/16	incorrect	gt	mlp_identity_scale	14	0.3651	[0.3177, 0.4134]
DeiT-S/16	incorrect	top1	mlp_identity_scale	14	0.3737	[0.3208, 0.4260]
DeiT-S/16	incorrect	margin	mlp_identity_scale	14	0.5210	[0.4896, 0.5517]
DeiT-S/16	incorrect	gt	mlp_output_identity_tv	14	-0.0829	[-0.1111, -0.0537]
DeiT-S/16	incorrect	top1	mlp_output_identity_tv	14	-0.0822	[-0.1229, -0.0378]
DeiT-S/16	incorrect	margin	mlp_output_identity_tv	14	-0.1514	[-0.1788, -0.1242]
DeiT-S/16	incorrect	gt	mlp_residual_path_cancellation	14	0.0745	[-0.0319, 0.1855]
DeiT-S/16	incorrect	top1	mlp_residual_path_cancellation	14	0.2091	[0.1068, 0.3061]
DeiT-S/16	incorrect	margin	mlp_residual_path_cancellation	14	-0.1185	[-0.1821, -0.0485]

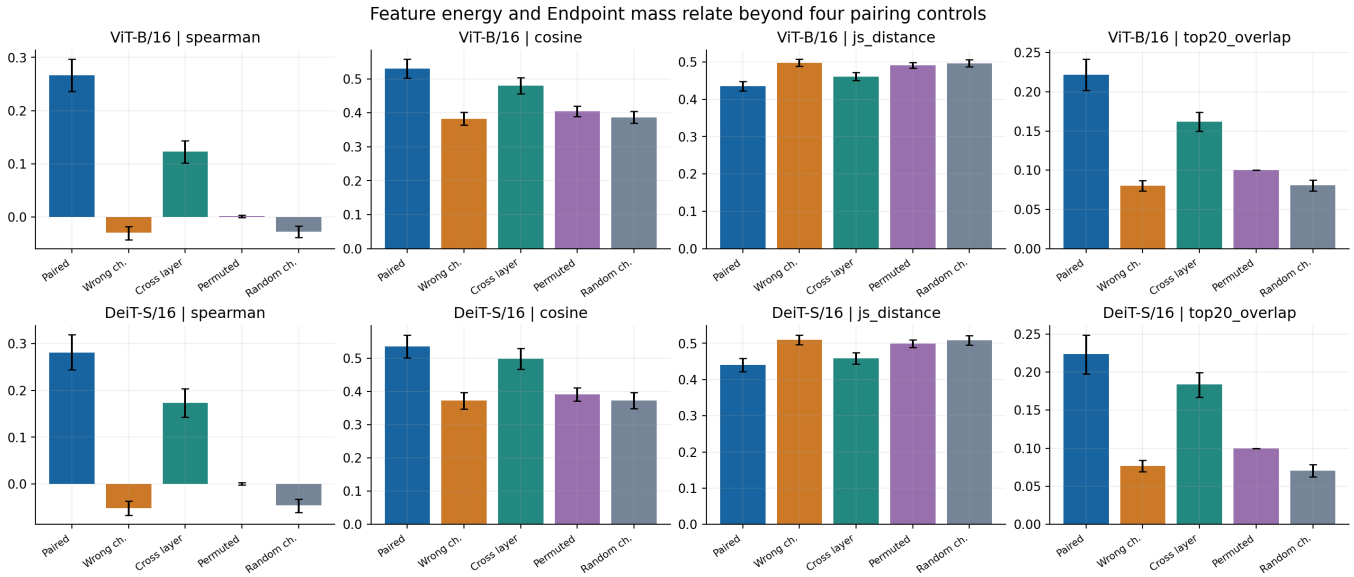
All block estimates, paired contrasts and NA counts · Image-block values

S12. Feature–Endpoint 定量关系

每模型 32 张冻结图像，12 blocks × 16 个冻结随机 channels；共 12,288 个 energy targets。求和覆盖 196 个 patch tokens，排除 CLS，与原前向展示中的 energy 定义一致。Endpoint 使用原始 RGB absolute mass 后聚合到 patch，未用展示归一化或空间方差选通道。

$$s_{b,c} = \frac{1}{2}\sum_{t=1}^{196} h_{b,t,c}^2, \qquad q_t = \frac{h_{b,t,c}^2}{\sum_{j=1}^{196} h_{b,j,c}^2}, \qquad p_t = \frac{\sum_{i\in P_t} |x_i \partial s_{b,c}/\partial x_i|}{\sum_i |x_i \partial s_{b,c}/\partial x_i|}. \tag{2}$$

正确配对的图像级 Spearman 均值为 0.2663/0.2811；与同层错误 channel、跨层同 channel、patch 置换和随机 channel 四种控制的配对差区间均高于零。Cosine、Jensen–Shannon distance 和 top-20 overlap 也逐项报告。结果表明同通道的前向 energy 与局部输入依赖具有超出这些控制的关联，关联强度仍远低于等价。图内先取 channel、block 中位数，再在图像间统计。



Model	Pairing	Metric	Mean	95% CI
ViT-B/16	paired	spearman	0.2663	[0.2366, 0.2973]

Model	Pairing	Metric	Mean	95% CI
ViT-B/16	wrong_channel	spearman	-0.0298	[-0.0426, -0.0177]
ViT-B/16	cross_layer	spearman	0.1232	[0.1019, 0.1435]
ViT-B/16	permuted	spearman	0.0013	[-0.0010, 0.0036]
ViT-B/16	random_channel	spearman	-0.0275	[-0.0389, -0.0175]
ViT-B/16	paired	cosine	0.5304	[0.5029, 0.5575]
ViT-B/16	wrong_channel	cosine	0.3826	[0.3643, 0.4011]
ViT-B/16	cross_layer	cosine	0.4802	[0.4562, 0.5032]
ViT-B/16	permuted	cosine	0.4043	[0.3888, 0.4198]
ViT-B/16	random_channel	cosine	0.3868	[0.3691, 0.4041]
ViT-B/16	paired	js_distance	0.4351	[0.4223, 0.4484]
ViT-B/16	wrong_channel	js_distance	0.4974	[0.4884, 0.5070]
ViT-B/16	cross_layer	js_distance	0.4604	[0.4498, 0.4716]
ViT-B/16	permuted	js_distance	0.4909	[0.4837, 0.4983]
ViT-B/16	random_channel	js_distance	0.4960	[0.4869, 0.5059]
ViT-B/16	paired	top20_overlap	0.2215	[0.2016, 0.2414]
ViT-B/16	wrong_channel	top20_overlap	0.0805	[0.0734, 0.0871]
ViT-B/16	cross_layer	top20_overlap	0.1621	[0.1496, 0.1742]
ViT-B/16	permuted	top20_overlap	0.1000	[0.1000, 0.1000]
ViT-B/16	random_channel	top20_overlap	0.0809	[0.0738, 0.0875]
DeiT-S/16	paired	spearman	0.2811	[0.2442, 0.3191]
DeiT-S/16	wrong_channel	spearman	-0.0514	[-0.0670, -0.0361]
DeiT-S/16	cross_layer	spearman	0.1735	[0.1430, 0.2036]
DeiT-S/16	permuted	spearman	0.0003	[-0.0021, 0.0029]
DeiT-S/16	random_channel	spearman	-0.0458	[-0.0611, -0.0318]
DeiT-S/16	paired	cosine	0.5359	[0.5010, 0.5697]
DeiT-S/16	wrong_channel	cosine	0.3723	[0.3477, 0.3966]
DeiT-S/16	cross_layer	cosine	0.4989	[0.4673, 0.5300]
DeiT-S/16	permuted	cosine	0.3912	[0.3718, 0.4111]
DeiT-S/16	random_channel	cosine	0.3733	[0.3490, 0.3974]
DeiT-S/16	paired	js_distance	0.4406	[0.4225, 0.4587]
DeiT-S/16	wrong_channel	js_distance	0.5094	[0.4962, 0.5228]
DeiT-S/16	cross_layer	js_distance	0.4584	[0.4429, 0.4740]
DeiT-S/16	permuted	js_distance	0.4990	[0.4886, 0.5098]
DeiT-S/16	random_channel	js_distance	0.5085	[0.4958, 0.5218]
DeiT-S/16	paired	top20_overlap	0.2234	[0.1973, 0.2484]
DeiT-S/16	wrong_channel	top20_overlap	0.0770	[0.0695, 0.0844]
DeiT-S/16	cross_layer	top20_overlap	0.1836	[0.1668, 0.1992]
DeiT-S/16	permuted	top20_overlap	0.1000	[0.1000, 0.1000]
DeiT-S/16	random_channel	top20_overlap	0.0707	[0.0625, 0.0785]

Model	Control	Metric	Paired similarity advantage	95% CI
ViT-B/16	wrong_channel	spearman	0.2961	[0.2763, 0.3165]
ViT-B/16	cross_layer	spearman	0.1431	[0.1292, 0.1579]
ViT-B/16	permuted	spearman	0.2649	[0.2350, 0.2961]
ViT-B/16	random_channel	spearman	0.2937	[0.2709, 0.3183]
ViT-B/16	wrong_channel	cosine	0.1478	[0.1367, 0.1584]
ViT-B/16	cross_layer	cosine	0.0502	[0.0429, 0.0582]
ViT-B/16	permuted	cosine	0.1261	[0.1126, 0.1391]
ViT-B/16	random_channel	cosine	0.1436	[0.1318, 0.1543]
ViT-B/16	wrong_channel	js_distance	0.0623	[0.0579, 0.0666]
ViT-B/16	cross_layer	js_distance	0.0253	[0.0220, 0.0286]
ViT-B/16	permuted	js_distance	0.0558	[0.0495, 0.0617]
ViT-B/16	random_channel	js_distance	0.0608	[0.0562, 0.0654]
ViT-B/16	wrong_channel	top20_overlap	0.1410	[0.1262, 0.1570]
ViT-B/16	cross_layer	top20_overlap	0.0594	[0.0484, 0.0711]
ViT-B/16	permuted	top20_overlap	0.1215	[0.1016, 0.1414]
ViT-B/16	random_channel	top20_overlap	0.1406	[0.1246, 0.1574]
DeiT-S/16	wrong_channel	spearman	0.3325	[0.3092, 0.3568]
DeiT-S/16	cross_layer	spearman	0.1076	[0.0975, 0.1181]
DeiT-S/16	permuted	spearman	0.2807	[0.2430, 0.3192]
DeiT-S/16	random_channel	spearman	0.3268	[0.3009, 0.3532]
DeiT-S/16	wrong_channel	cosine	0.1636	[0.1527, 0.1741]

Model	Control	Metric	Paired similarity advantage	95% CI
DeiT-S/16	cross_layer	cosine	0.0370	[0.0315, 0.0425]
DeiT-S/16	permuted	cosine	0.1447	[0.1259, 0.1633]
DeiT-S/16	random_channel	cosine	0.1626	[0.1507, 0.1740]
DeiT-S/16	wrong_channel	js_distance	0.0687	[0.0638, 0.0737]
DeiT-S/16	cross_layer	js_distance	0.0177	[0.0146, 0.0207]
DeiT-S/16	permuted	js_distance	0.0584	[0.0501, 0.0663]
DeiT-S/16	random_channel	js_distance	0.0679	[0.0628, 0.0730]
DeiT-S/16	wrong_channel	top20_overlap	0.1465	[0.1262, 0.1660]
DeiT-S/16	cross_layer	top20_overlap	0.0398	[0.0297, 0.0504]
DeiT-S/16	permuted	top20_overlap	0.1234	[0.0973, 0.1484]
DeiT-S/16	random_channel	top20_overlap	0.1527	[0.1328, 0.1723]

Unit metrics · Image-block values

S13. 抽样敏感性

在相同 32 图上复用 M3 原 32 channels 和 15 patch strata。channel 与 token-stratum bootstrap 各 256 次，image bootstrap 2,000 次，并逐一移除 block 或空间 stratum。下表给出重采样分布和删除敏感性范围。前三个空间趋势方向在所有敏感性范围内保持一致。该分析没有生成独立 channel-seed 复现实验。CLS−patch 列为全层差值；其余为 late−early。

Model	Metric	Reference mean	Image CI	Channel bootstrap range	Token bootstrap range	Leave-one-block range	Leave-one-stratum range
ViT-B/16	entropy	0.2392	[0.2303, 0.2480]	[0.2307, 0.2472]	[0.2326, 0.2421]	[0.1847, 0.2937]	[0.2375, 0.2403]
ViT-B/16	locality	-0.3607	[-0.3683, -0.3526]	[-0.3678, -0.3523]	[-0.3641, -0.3526]	[-0.4091, -0.3123]	[-0.3617, -0.3596]
ViT-B/16	distance	0.1331	[0.1274, 0.1389]	[0.1295, 0.1358]	[0.1168, 0.1478]	[0.1182, 0.1479]	[0.1296, 0.1366]
ViT-B/16	cancellation	0.0024	[0.0021, 0.0028]	[0.0020, 0.0029]	[0.0022, 0.0027]	[0.0016, 0.0032]	[0.0024, 0.0025]
ViT-B/16	cls_minus_patch_entropy	0.0564	[0.0454, 0.0669]	[0.0551, 0.0590]	[0.0527, 0.0636]	[0.0528, 0.0600]	[0.0548, 0.0580]
DeiT-S/16	entropy	0.2236	[0.2167, 0.2299]	[0.2190, 0.2279]	[0.2185, 0.2274]	[0.1800, 0.2673]	[0.2232, 0.2254]
DeiT-S/16	locality	-0.3412	[-0.3527, -0.3300]	[-0.3447, -0.3376]	[-0.3494, -0.3315]	[-0.3797, -0.3026]	[-0.3429, -0.3378]
DeiT-S/16	distance	0.1261	[0.1191, 0.1338]	[0.1249, 0.1277]	[0.1138, 0.1381]	[0.1176, 0.1347]	[0.1239, 0.1304]
DeiT-S/16	cancellation	0.0020	[0.0016, 0.0025]	[0.0018, 0.0022]	[0.0018, 0.0024]	[0.0014, 0.0025]	[0.0019, 0.0021]
DeiT-S/16	cls_minus_patch_entropy	0.0829	[0.0699, 0.0943]	[0.0813, 0.0850]	[0.0741, 0.0931]	[0.0749, 0.0909]	[0.0794, 0.0848]

S14. 跨层全-head M2 与适用范围

主文 block-5 三-head 实验对应原确认性协议。已封存 M2 覆盖 blocks 2/5/10、两模型、全部 heads、两种 references、每 family 400 图；原生 patch 几何下完整输入替换 11/12、单-head 隔离替换 12/12 显著偏向 regional Taylor。下表使用原全部 672 个双向 AUC 检验的 Holm 校正，未重新定义检验 family。

Model	Reference	Block	Outcome	n images	Taylor > Endpoint Holm p	Supported
DeiT-S/16	blur	10	full	400	5.841e-20	True
DeiT-S/16	blur	10	isolated	400	2.355e-82	True
DeiT-S/16	normalized-zero	10	full	400	0.0282	True
DeiT-S/16	normalized-zero	10	isolated	400	4.045e-90	True
DeiT-S/16	blur	2	full	400	1.103e-08	True
DeiT-S/16	blur	2	isolated	400	6.597e-69	True
DeiT-S/16	normalized-zero	2	full	400	0.04217	True
DeiT-S/16	normalized-zero	2	isolated	400	2.276e-58	True
DeiT-S/16	blur	5	full	400	6.793e-15	True

Model	Reference	Block	Outcome	n images	Taylor > Endpoint Holm p	Supported
DeiT-S/16	blur	5	isolated	400	7.52e-85	True
DeiT-S/16	normalized-zero	5	full	400	0.5283	False
DeiT-S/16	normalized-zero	5	isolated	400	2.355e-82	True
ViT-B/16	blur	10	full	400	5.562e-28	True
ViT-B/16	blur	10	isolated	400	2.351e-94	True
ViT-B/16	normalized-zero	10	full	400	2.309e-26	True
ViT-B/16	normalized-zero	10	isolated	400	2.424e-97	True
ViT-B/16	blur	2	full	400	4.341e-16	True
ViT-B/16	blur	2	isolated	400	6.185e-64	True
ViT-B/16	normalized-zero	2	full	400	2.49e-17	True
ViT-B/16	normalized-zero	2	isolated	400	7.981e-47	True
ViT-B/16	blur	5	full	400	2.591e-25	True
ViT-B/16	blur	5	isolated	400	1.679e-91	True
ViT-B/16	normalized-zero	5	full	400	9.405e-18	True
ViT-B/16	normalized-zero	5	isolated	400	1.042e-115	True

本轮未执行可选 cascading randomization、P2 同目标 M3–M4 桥接、独立 channel seeds、替代 checkpoint，或 P3 输入变换与 CNN 展示。本文不把训练对照解释为有限因果机制，也不把不同标量目标之间的总体关联解释为逐单元桥接。Path IG 的原冻结校准失败及原确认性结论保持不变。

Frozen protocol · Source bindings · Task acquisition · Numerical validation

S15. 数据、模型、总体与分析身份

ImageNet 来源为 ILSVRC2012 官方验证集。模型由 timm 0.9.16 标准 Block/Attention 实例化并严格加载权重，所有权重文件已重新核验大小和 SHA-256。AugReg 名称标明 IN21K 预训练后 IN1K 微调或 IN1K 训练；DeiT-S 使用公开 Facebook IN1K 参数。所有结果条件于这些参数状态。

Model ID	Exact timm name / training	Bytes	SHA-256
vit_b16	vit_base_patch16_224.augreg2_in21k	346329983	489bfa2f1aa3b6781e268a87eb629816b89eebba43942
deit_s16	deit_small_patch16_224.fb_in1k	88251143	cd65a15597004d0ce19d7a9daef969903972db5b398e3
vit_b16_in1k	vit_base_patch16_224.augreg_in1k	346335542	69c9a2822a47838f57e522350b10ccc0d89d67fdcee076
vit_s16_augreg	vit_small_patch16_224.augreg_in21k	88267574	545815b4e770d2fa6ca4b3ccba7c16b035e474354e52d

原 discovery、calibration、evaluation 分别 64、64、400 类，每类一图。Unit-study 的 100 类各一图是 evaluation 子集。E1 为 100 类各五图，E5 为 400 类各一图。E5 从排除历史曝光和 E1 的候选集中按 class_e5/image_e5 哈希排序选择。以下矩阵报告图像身份重叠；同类别可以跨后续队列出现，图像独立不等于类别独立。E5 的 train/validation/test 标签用于另一个预测分析，不改变此处全部 400 图 H1–H4 检查。

Cohort	discovery	calibration	evaluation	unit_study	e1	e5
discovery	64	0	0	0	0	0
calibration	0	64	0	0	0	0
evaluation	0	0	400	100	0	0
unit_study	0	0	100	100	0	0
e1	0	0	0	0	500	0
e5	0	0	0	0	0	400

预处理使用 Pillow 10.2.0 RGB，不应用 EXIF transpose。Torchvision 0.17.2 先对 PIL 图像以 bicubic 将短边 resize 到 248，antialias=True，再中心 crop 224，转 float32 并除 255。每模型 normalization 使用 Table 2 参数。Blur 在 crop 后 RGB 上使用 31×31 Gaussian kernel、sigma=10、reflect padding=15，再 normalization。Evaluation mode 关闭 dropout 和 drop path。CLS、位置编码与 bias 固定。原 logit 目标为官方 ground-truth，clean-top1 为另一个预先

固定读出；它们不参与 cohort 选择。原 registry 每层四 tokens/四 channels 的规则和实际列表由封存 registry 给出。M3 固定 CLS 加 15 个 3×5 分层 tokens，每层 32 channels；跨模型同序号不代表同一语义神经元。结构研究扫描全部 heads 和 12 blocks，区域主任务为 block 5，ViT-B heads 0/6/11、DeiT-S heads 0/3/5。

所有抽样是确定性规则或公开种子。实际 selected IDs 与 targets 由 cohort bindings 提供。随机控制的具体派生键与调用顺序保留于原协议，不能把不同分析的种子混用。

Purpose	Frozen seed / deterministic rule
Original image selection	SHA256(NNS_VIT_ENDPOINT_EP0_R1_20260918 | class | WNID), then namespace | image | filename
Unit cohort	SHA256(ENDPOINT_M3_COHORT_20260919 | sample_id | image_sha256), lowest 100
Unit channels	SHA256(ENDPOINT_M3_CHANNEL_20260919 | model | block | channel), lowest 32
Unit tokens	CLS + nearest patch to each of 3×5 strata centers; row-major ties
Original bootstrap	SHA256(namespace | bootstrap | family_id), first 8 bytes big-endian; PCG64
Unit bootstrap	SHA256(ENDPOINT_M3_BOOTSTRAP_20260919 | M3 | model | metric), first 8 bytes
Structural bootstrap	SHA256(M4 | test | model | test_id), with residual type inserted where applicable
Random initialization	202609201, 202609202, 202609203
Independent cohort / target selection	NNS_VIT_REVISION_V5_20260921_R1 namespace; class_e5, image_e5 and post/block/channel/token keys
Independent-cohort bootstrap	ENDPOINT_M1_20260918_V1 | 20260918 | bootstrap | V5 | (model,hypothesis)
Coordinate/threshold bootstrap	SHA256(2026092302 | analysis key), first 8 bytes; 10,000 PCG64 replicates
Analytic directions	NumPy default_rng(2026092301), three normalized Gaussian directions
Regional random orders	Frozen namespace/model/image/reference/head/repeat keys in original intervention registry; 20 repeats

确认性身份相对于各自分析前的本地冻结计划。外部公开注册未提供。下表保留各 family 的原成员与用途；后续检查不与原 28 项追溯合并。完整 28 个 test IDs 与所有 SHA-256 位于 analysis identity。0.10/0.05 没有外部效用校准来源，属于研究者操作阈值。原判断要求有效性、完整覆盖、n 和区间精度门，以及点估计和 Holm 检验共同通过；不要求 CI lower > 0.10。阈值敏感性见 68 行完整结果。

Analysis	Identity	Multiplicity family	Local freeze evidence
original_component_and_region	confirmatory	28 tests jointly Holm; 12 component correlations + 16 regional baseline contrasts	local pre-outcome plan; first committed occurrence below; 6237ef87a1af0205973b76bee172defe33a0a6da 2026-09-18T09:26:20+08:00
unit_depth	confirmatory	five per model: entropy, locality, distance, cancellation, CLS-patch entropy	local pre-outcome plan; first committed occurrence below; f2eba3f60d765e6a9729d9ff12a57bd01f93f952 2026-09-19T13:53:48+08:00
attention_residual_depth	confirmatory	four attention tests/model; two residual tests/model/type	local pre-outcome plan; first committed occurrence below; 1a1000bcdd15d8b0e76406f4376a82d60ec79b86 2026-09-19T17:25:17+08:00
head_baselines	exploratory	separate paired and absolute Spearman families; other metrics descriptive	local pre-outcome plan; first committed occurrence below; 33c8c320446a631b2e451016b3268eefdd9c926b 2026-09-19T04:20:36+08:00
residual_spatial_tv	post-hoc	four Holm contrasts across model and residual type	2026-09-19T10:59:17.569991+00:00; protocol timestamp and hashes
QKV_spatial	prospective robustness	30 descriptive contrasts; no confirmatory family combined with prior tests	2026-09-20T17:14:57.250515+00:00; protocol timestamp and hashes

Analysis	Identity	Multiplicity family	Local freeze evidence
independent_cohort_checkpoints	prospective robustness	H1-H4 Holm separately within each checkpoint	local pre-outcome plan; first committed occurrence below; 944f54c99c8d83803d461224565b93a779e75915 2026-09-21T11:31:10+08:00
training_target_sampling_controls	exploratory	descriptive 2000 bootstrap; training control resamples image and initialization seed; sampling control resamples target strata and images	2026-09-20T00:22:59.620440+00:00; 395d30b3804983d0e51e99a28d0c3c900aca8fda 2026-09-20T22:41:48+08:00
coordinate_sensitivity_v6	prospective robustness	no significance-based success criterion; report all directions and intervals	2026-09-22T16:43:58.690139+00:00; protocol timestamp and hashes
threshold_sensitivity_v6	post-hoc	descriptive thresholds only; original tests and decisions immutable	2026-09-22T16:43:58.690139+00:00; protocol timestamp and hashes

S16. 有限替换变量与可执行伪代码

A 和 V 是前向变量，routing-induced/value-induced 是微分分组。Table 6 的 A/V family 每个 block 各一个全-head A 替换和 V 替换，共 24 成分；head family 为所有 block/head，residual family 每 block attention 与 MLP branch 各一项。Reference activation 来自完整 reference forward。下面的 consumer slot 是替换身份的一部分。

Replacement	Forward tensor / shape (batch=1)	Consumer input slot	Clean state and recomputation
Attention probability	A [1,H,197,197], after softmax/dropout	left input of A @ V	V, other branches and residual identity clean; recompute AV, concat, output projection, residual suffix
V projection	V [1,H,197,64], qkv projection slice	right input of A @ V	Q/K/A and identity clean; recompute AV and entire suffix
Head output	slice of projection_input [1,197,H×64], head h channels h×64:(h+1)×64	attn.proj input channels	all other head slices and residual identity clean; recompute projection and suffix
Attention branch	drop_path1 output [1,197,D]	residual1 add branch input	identity u clean; recompute residual sum, MLP and remaining blocks
MLP branch	drop_path2 output [1,197,D]	residual2 add branch input	carry r1 clean; recompute residual sum and remaining blocks

```
clean_rgb = RGB(image) / 255
clean_rgb = center_crop(resize_shorter_side(clean_rgb, 248, bicubic, antialias=True), 224)
x = (clean_rgb - mu) / sigma
reference_zero = zeros_like(x)              # raw RGB equals mu
reference_blur = (GaussianBlur(clean_rgb, kernel=31, sigma=10, reflect_padding=15) - mu) / sigma
clean = forward_and_capture(x)
reference = forward_and_capture(reference_input)
for fixed block, component, consumer slot:
    replacement = reference.tensor_at(component, slot)
    patched = forward(x, replace_only_this_use=replacement)
    effect = clean.GT_logit - patched.GT_logit
    # All sibling heads and bypass inputs remain clean before the replacement.
    # Every node downstream of the replaced use is recomputed normally.

for singleton region P in the fixed 4x4 grid of 56x56 pixels:
    x_P = x.copy(); x_P[P] = reference_input[P]
    mixed[P] = forward_and_capture(x_P)
```

```
    regional_Taylor[P] = abs(dot(clean_head_cotangent, clean_head - mixed[P].head))
rank_regions(score, ties=ascending_region_id)
for budget in [1, 2, 4]:
    selected = top_regions(budget)
    x_selected = x with selected pixels from reference_input
    mixed = forward_and_capture(x_selected)
    full_input_effect = clean.GT_logit - mixed.GT_logit
    isolated = forward(x, target_head_slice=mixed.head_slice)
    isolated_effect = clean.GT_logit - isolated.GT_logit
AUC = trapezoid(abs(isolated_effect), areas=[0,1/16,2/16,4/16]) / 0.25
image_AUC = mean(AUC across the three fixed heads)
random_AUC = mean(AUC per random repeat, then heads) # abs before averaging repeats
```

Full-input effect 改变整条前向计算；isolated-component effect 只将 mixed forward 的目标张量放入 clean forward，其他 head 与 bypass 保持 clean。区域分数的候选集合相同，但 Taylor 需额外 singleton mixed forwards，因此信息预算不同。区域 ties 用编号升序，head ties 用 component ID 升序，Spearman 用平均秩。所谓 candidate-envelope AUC 是在每个面积点从已评价候选中取最大 absolute effect，再积分、平均 heads，可能拼接不同排序的点，因此不是一条可执行的全局最优区域策略。它只用作已评价集合的上界和 regret 分母。

S17. 坐标敏感性与独立解析检查

三种表示均在同一当前输入点、同一固定目标与 checkpoint 上计算。Raw gradient 除去模型 sigma；common-origin weighted gradient 使用原始 RGB 零点，不随模型 mean 改变。以下每格为图像级中位差和 95% bootstrap CI。Unit 每模型 n=100，结构项 n=400；缺失图像为零。每个 coordinate 在图内配对，不作为额外样本。

$$E_i^{(x)} = x_i \frac{\partial s}{\partial x_i}, \qquad G_i^{(r)} = \frac{1}{\sigma_i}\frac{\partial s}{\partial x_i}, \qquad E_i^{(0)} = r_i \frac{\partial s}{\partial r_i} = \left(x_i + \frac{\mu_i}{\sigma_i}\right)\frac{\partial s}{\partial x_i}. \tag{3}$$

Model	Object / metric	Contrast	Normalized Endpoint	Raw-pixel gradient	Common-origin gradient	n / NA
deit_s16	K / entropy	block11_minus_block0	-0.067 [-0.074, -0.059]	-0.066 [-0.073, -0.060]	-0.061 [-0.069, -0.053]	400 / 0
deit_s16	K / entropy	late_minus_early	-0.028 [-0.031, -0.024]	-0.029 [-0.032, -0.026]	-0.027 [-0.031, -0.024]	400 / 0
deit_s16	Q / entropy	block11_minus_block0	-0.004 [-0.008, 0.0005]	0.003 [-0.001, 0.006]	0.002 [-0.003, 0.006]	400 / 0
deit_s16	Q / entropy	late_minus_early	0.018 [0.013, 0.022]	0.018 [0.015, 0.022]	0.015 [0.012, 0.018]	400 / 0
deit_s16	V / entropy	block11_minus_block0	-0.067 [-0.075, -0.061]	-0.076 [-0.081, -0.068]	-0.070 [-0.077, -0.062]	400 / 0
deit_s16	V / entropy	late_minus_early	-0.013 [-0.016, -0.011]	-0.016 [-0.017, -0.013]	-0.015 [-0.017, -0.013]	400 / 0
deit_s16	attention / output_identity_tv	block11_minus_block0	0.078 [0.072, 0.085]	0.079 [0.070, 0.085]	0.074 [0.067, 0.081]	400 / 0
deit_s16	attention / output_identity_tv	late_minus_early	0.028 [0.026, 0.030]	0.028 [0.026, 0.030]	0.027 [0.026, 0.029]	400 / 0
deit_s16	mlp / output_identity_tv	block11_minus_block0	-0.050 [-0.059, -0.040]	-0.044 [-0.057, -0.037]	-0.037 [-0.043, -0.029]	400 / 0
deit_s16	mlp / output_identity_tv	late_minus_early	-0.046 [-0.052, -0.041]	-0.044 [-0.051, -0.039]	-0.039 [-0.044, -0.034]	400 / 0
deit_s16	unit / distance	block11_minus_block0	0.163 [0.159, 0.169]	0.169 [0.164, 0.175]	0.171 [0.167, 0.178]	100 / 0
deit_s16	unit / distance	late_minus_early	0.122 [0.120, 0.128]	0.129 [0.126, 0.132]	0.127 [0.122, 0.131]	100 / 0
deit_s16	unit / entropy	block11_minus_block0	0.397 [0.387, 0.404]	0.426 [0.412, 0.438]	0.418 [0.400, 0.427]	100 / 0
deit_s16	unit / entropy	late_minus_early	0.225 [0.212, 0.228]	0.245 [0.241, 0.249]	0.221 [0.217, 0.229]	100 / 0

Model	Object / metric	Contrast	Normalized Endpoint	Raw-pixel gradient	Common-origin gradient	n / NA
deit_s16	unit / locality	block11_minus_block0	-0.455 [-0.467, -0.432]	-0.453 [-0.469, -0.441]	-0.477 [-0.493, -0.455]	100 / 0
deit_s16	unit / locality	late_minus_early	-0.342 [-0.353, -0.333]	-0.359 [-0.366, -0.352]	-0.356 [-0.361, -0.344]	100 / 0
vit_b16	K / entropy	block11_minus_block0	-0.059 [-0.064, -0.053]	-0.052 [-0.058, -0.048]	-0.046 [-0.054, -0.041]	400 / 0
vit_b16	K / entropy	late_minus_early	-0.021 [-0.024, -0.019]	-0.019 [-0.023, -0.016]	-0.019 [-0.022, -0.016]	400 / 0
vit_b16	Q / entropy	block11_minus_block0	0.027 [0.022, 0.031]	0.026 [0.018, 0.031]	0.014 [0.009, 0.018]	400 / 0
vit_b16	Q / entropy	late_minus_early	0.037 [0.033, 0.043]	0.039 [0.033, 0.044]	0.034 [0.030, 0.037]	400 / 0
vit_b16	V / entropy	block11_minus_block0	-0.053 [-0.059, -0.047]	-0.057 [-0.061, -0.049]	-0.055 [-0.060, -0.050]	400 / 0
vit_b16	V / entropy	late_minus_early	-0.010 [-0.012, -0.008]	-0.009 [-0.011, -0.007]	-0.010 [-0.012, -0.007]	400 / 0
vit_b16	attention / output_identity_tv	block11_minus_block0	0.166 [0.159, 0.172]	0.167 [0.159, 0.174]	0.167 [0.160, 0.175]	400 / 0
vit_b16	attention / output_identity_tv	late_minus_early	0.052 [0.049, 0.055]	0.054 [0.051, 0.056]	0.053 [0.050, 0.055]	400 / 0
vit_b16	mlp / output_identity_tv	block11_minus_block0	-0.037 [-0.043, -0.031]	-0.034 [-0.038, -0.029]	-0.032 [-0.039, -0.025]	400 / 0
vit_b16	mlp / output_identity_tv	late_minus_early	-0.019 [-0.022, -0.015]	-0.017 [-0.020, -0.013]	-0.018 [-0.021, -0.014]	400 / 0
vit_b16	unit / distance	block11_minus_block0	0.197 [0.190, 0.204]	0.205 [0.200, 0.211]	0.209 [0.203, 0.214]	100 / 0
vit_b16	unit / distance	late_minus_early	0.131 [0.129, 0.136]	0.138 [0.134, 0.141]	0.134 [0.129, 0.138]	100 / 0
vit_b16	unit / entropy	block11_minus_block0	0.398 [0.392, 0.409]	0.440 [0.426, 0.447]	0.412 [0.402, 0.426]	100 / 0
vit_b16	unit / entropy	late_minus_early	0.220 [0.211, 0.227]	0.240 [0.235, 0.246]	0.217 [0.209, 0.223]	100 / 0
vit_b16	unit / locality	block11_minus_block0	-0.526 [-0.534, -0.516]	-0.543 [-0.554, -0.532]	-0.545 [-0.552, -0.538]	100 / 0
vit_b16	unit / locality	late_minus_early	-0.356 [-0.363, -0.350]	-0.367 [-0.373, -0.361]	-0.356 [-0.364, -0.347]	100 / 0

DeiT-S Q 的 B11−B0 在三坐标下的点估计符号不同且三个区间跨零。Phase contrast 与 B11−B0 的差异还说明，终点比较不能替代整个轨迹。主文始终限定 normalized-input Endpoint。其他对象只作这三表示下的方向保持判断，不推出一般参数化不变性。

独立参考以 NumPy float64 显式构造 affine、LayerNorm、softmax $\mathrm{diag}(a) - aa^{\mathsf{T}}$、Q/K/V、head concat/output projection、GELU/MLP 和 residual identity Jacobian。参考模块不导入 scanner、hook 或 autograd。独立驱动将其结果与实际主扫描器在单头、双头、pre-LN residual attention/MLP tiny 模型上逐项比较。32 个分量比较和 9 个冻结方向有限差分通过，最大分量绝对误差 1.11×10^{-16}；每项 relative L2 和容差见 analytic results。共享 producer 双 slot toy、Q/K slot 交换、提前合并 fan-out 和重复合并项均被拒绝，正确实现先通过。此能力限于这些数值图和故障类型，不等于排除全部 scanner 实现错误。

原区域数组键 raw_gradient 记录 $\nabla_x s$，不是 $\nabla_r s$。主文已按实际坐标重命名。Random region order 以 NNS_VIT_ENDPOINT_E5_RANDOM_REGIONS_V1、task_seed、model_id、reference_id、head、repeat 拼接字符串作 SHA-256，前八字节 big-endian 作为 PCG64 seed，再 permutation(16)。task_seed 来自 NNS_VIT_ENDPOINT_QUEUE_V1 与 task_id。

S18. Revision v7 分类型、层级与质量诊断

本节结果来自现役 nns_vit_endpoint_revision_v7_r4_20260923 CPU-only 根。R1–R3 的统计或 provenance 问题按递增根保留；R4 恢复 R2 冻结的 bootstrap seed，限定 concordance 为描述性比例，在 residual-TV 质量分层中先按

图聚合，并在执行前绑定最终 analysis/validator/test 哈希。旧 pooled family、28 项 Holm 和全部 sealed 结论均未改变。

Table S18.1. 分类型相关；图像级中位数 [95% CI]。每项在一张图的 12 个 blocks 间计算；这是事后诊断。

Model	Reference	A only	V only	Attention branch only	MLP branch only
vit_b16	normalized-zero	0.049 [0.007, 0.091]	0.224 [0.192, 0.259]	-0.028 [-0.056, -0.003]	0.287 [0.266, 0.318]
vit_b16	blur	0.073 [0.028, 0.112]	0.311 [0.266, 0.350]	0.000 [-0.021, 0.042]	0.378 [0.336, 0.402]
deit_s16	normalized-zero	0.091 [0.049, 0.119]	0.035 [-0.021, 0.087]	-0.157 [-0.189, -0.119]	0.378 [0.346, 0.406]
deit_s16	blur	0.101 [0.056, 0.133]	0.147 [0.112, 0.182]	-0.024 [-0.070, 0.014]	0.371 [0.332, 0.395]

Table S18.2. 类型/层级残差化 rank correlation 与 head 内相关。O、B 和 O+B 分别表示控制 operation、block、二者；逐图先对 m 与 $|e|$ 取平均秩，再以带截距的 one-hot 固定效应最小二乘残差化。Head 列先在每层 heads 内计算 Spearman，截断至 $[-1+10^{-12}, 1-10^{-12}]$ 后作 Fisher-z 平均并反变换。

Model	Reference	A/V O	A/V B	A/V O+B	Residual O	Residual B	Residual O+B	Head within-block	A/V concordance
vit_b16	normalized-zero	0.122	-0.044	0.030	0.168	-0.010	-0.128	0.321	0.500
vit_b16	blur	0.164	-0.015	-0.018	0.184	0.134	0.042	0.303	0.500
deit_s16	normalized-zero	0.077	0.167	0.089	0.110	0.019	-0.091	0.245	0.583
deit_s16	blur	0.139	0.186	0.095	0.161	0.280	0.142	0.183	0.583

A/V concordance 是 12 层中 $\text{sign}[(m_A - m_V)(|e_A| - |e_V|)]$ 一致的比例；零乘积作为 tie 排除。它不是 Spearman，仅作描述性报告，不做以 0 为原假设的符号检验。完整置信区间、分位数、单层结果、自由度、condition number 与独立验证位于 v7 r4 根。所有必需相关都有 400/400 个有效图像。

结果说明旧 pooled family 的正相关不能解释为类型内部统一排序能力。V-only 在 ViT-B 较强，DeiT-S normalized-zero 的区间跨零；MLP branch 在四种设置中为正，attention branch 为负或接近零。类型和 block 同时控制后，A/V 在 ViT-B 两种 reference 下接近零，residual 在 normalized-zero 下为负；DeiT-S 的调整结果随 reference 改变。

Table S18.3. Own-token locality 半径敏感性；late-minus-early 中位差 [95% CI]。

Model	Radius 0	Radius 1	Radius 2
vit_b16	-0.302 [-0.310, -0.298]	-0.356 [-0.363, -0.350]	-0.335 [-0.344, -0.322]
deit_s16	-0.260 [-0.263, -0.253]	-0.342 [-0.353, -0.333]	-0.315 [-0.323, -0.303]

三个半径的方向一致。半径 1 仍是预先定义的主描述量；该敏感性不用于选择新主指标。

Residual output–identity TV 的 absolute-mass 四分位诊断以图像为统计单位。Attention 的 Q1→Q4 中位 TV 为 ViT-B 0.107→0.073、DeiT-S 0.071→0.055；MLP 为 ViT-B 0.139→0.105、DeiT-S 0.117→0.090。空间差异不只存在于最低质量组，但这些未调整的事后分层不能排除梯度质量或噪声的影响。

H4 从 1,600 个原始 JSON/NPZ 独立复算；四个 checkpoint 的中位数依次为 vit_b16 -0.114、deit_s16 -0.186、vit_b16_in1k -0.186、vit_s16_augreg -0.136，正/负/平分别为 101/297/2、96/298/6、56/342/2、107/285/8。四项 checkpoint 内 Holm 调整值均小于 10^{-6}。原始逐图、block-level 相关、完整 p 值及机器收据见 v7 r4 根。

S19. 架构切面、空间熵质量与 H4 聚合敏感性

S19.1 三类 registry 与实际切面认证

机器合同现按对象分为标量目标、局部微分分解和有限替换干预三个只读派生 registry。标量目标数由原 EP0 逐行重算，ViT-B/16 与 DeiT-S/16 分别为 457/385。每个模型均含 192 个 post-block token-channel、48 个 patch sum、48 个 patch energy、18 个 module patch aggregate、4 个 patch-embedding aggregate、1 个 final-CLS coordinate 和 2 个类别 logits；head coordinate 分别为 144 与 72。Q/K/V-induced、routing/value、residual 分项和可替换变量不再混入这一计数。

Registry	包含对象	明确不包含
scalar target	标量 readout、图中位置、聚合规则	路径分项、干预变量
local-differential decomposition	native use、解析分组、cut、重构对象	标量目标数量、有限替换效果
finite-replacement intervention	可替换 activation、reference、effect 与区域任务	局部微分闭合对象

冻结的 pre-LN ViT tensor-use DAG 对 842 个 target、12 层与 8 类 cut 的 80,832 个配对逐一检查。算法在目标祖先子图上动态传播每条输入路径经过的 cut-edge 数，并要求完整 cut 的计数集合严格为 $\{1\}$，同时要求 cut 中每条边均属于该祖先子图。26,094 个配对通过完整 target-cut 条件，15,660 个通过 attention branch-only partition 条件，24,444 个因声明重构对象不适用而拒绝，14,634 个因目标位于上游或 branch 不在目标祖先图而拒绝。该证明覆盖 registry 绑定的标准 unfused pre-LN ViT；数值闭合与 mutation 测试继续作为实现层面的独立证据。

S19.2 空间熵的总质量敏感性

阈值在每个 model–block 内由全部冻结 M3 unit map 的总 L1 mass 分布确定。原始 late-minus-early entropy 从源数组复现的最大误差为 0。下表以图像为单位给出中位数和 95% bootstrap 区间；截尾项保留全部 100 图，四分位项对合法空组传播 NA。

Model	Analysis	Median [95% CI]	Valid / total
ViT-B/16	no trimming	0.220 [0.211, 0.227]	100 / 100
ViT-B/16	trim lowest 1% mass	0.222 [0.213, 0.230]	100 / 100
ViT-B/16	trim lowest 5% mass	0.229 [0.221, 0.237]	100 / 100
ViT-B/16	trim lowest 10% mass	0.234 [0.227, 0.242]	100 / 100
DeiT-S/16	no trimming	0.225 [0.212, 0.228]	100 / 100
DeiT-S/16	trim lowest 1% mass	0.226 [0.215, 0.229]	100 / 100
DeiT-S/16	trim lowest 5% mass	0.230 [0.219, 0.235]	100 / 100
DeiT-S/16	trim lowest 10% mass	0.233 [0.223, 0.237]	100 / 100

按 model–block mass 四分位分层后，ViT-B 的 Q1–Q4 中位差为 0.090、0.194、0.238、0.266，DeiT-S 为 0.112、0.205、0.222、0.224。八个 image–quartile 组合因 early 或 late 组为空而记为 typed NA，没有删除或替换图像。同图同层 log1p(total mass) 与 entropy 的平均 Spearman 中位数为 ViT-B -0.729 [-0.747, -0.693]、DeiT-S -0.760 [-0.775, -0.723]；late-minus-early log1p mass 则分别为 2.893 [2.829, 2.961] 与 2.183 [2.109, 2.227]。因此熵与质量明显相关，但总体扩展方向不由最低质量图单独产生。

S19.3 H4 匹配聚合与随机区域精度

匹配聚合将 path score 定义为 $M^{\mathrm{sum}}_{jbh} = |\sum_i \mathcal{L}_{jbh,i}|$，使其与 activation × gradient 对照都采用 “先坐标求和、再取绝对值”。其余 target、normalized-zero reference、四个 blocks、有限替换 effect 与图像级聚合不变。

Checkpoint	Matched path minus activation × gradient	Positive / negative / tie
vit_b16	−0.124 [−0.147, −0.102]	89 / 311 / 0
deit_s16	−0.114 [−0.157, −0.079]	130 / 261 / 9
vit_b16_in1k	−0.226 [−0.253, −0.205]	49 / 349 / 2
vit_s16_augreg	−0.171 [−0.200, −0.143]	110 / 285 / 5

原 H4 从相同数组再次复现，逐图最大误差为 0。匹配聚合下四个 checkpoint 仍为负，因而原结果不由两种分数的 absolute/sum 顺序差异解释。

随机区域基线无需新增模型运行。源复核覆盖 1,600 个 model–reference–image 条件和 32,000 条 repeat 记录；每个条件均有 20 个完整重复。相对 Monte Carlo 标准误中位数为 5.1%–6.9%，q90 为 7.6%–8.9%；5/10/20-repeat 重采样、split-half 与 leave-one-repeat-out 均已保留。该精度支持 cohort-level 比较，单图细小优势不据此判定。

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